

# Child Protection in Response to Public Health Crises: Evidence from the Opioid Epidemic

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## Abstract

This paper examines the impacts of the opioid epidemic on child maltreatment and how Child Protective Services (CPS) responded to the resulting increase in maltreatment risk. I exploit the reformulation of OxyContin, one of the largest opioid supply disruptions in U.S. history that exacerbated the opioid epidemic. Leveraging variation in states' and counties' exposure to this nationwide supply shock, I find that the opioid epidemic led to an increase in maltreatment allegations, suggesting a rise in the underlying maltreatment risk. Despite heightened risk, institutional responses did not adjust accordingly, resulting in more cases where children were left at home but subsequently re-reported for maltreatment. Foster care placement rates remained unchanged and child welfare expenditures from federal sources per allegation declined, pointing to a limited systemic response. This study underscores the intergenerational consequences of the opioid epidemic and highlights how public health crises can weaken protective systems, particularly when institutions lack the flexibility or capacity to adapt.

**Keywords:** Opioid Epidemic, Child Maltreatment, Foster Care, Child Protective Services

**JEL Codes:** I12, I18, J13, J18

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# 1 Introduction

Public health crises have widespread consequences, extending beyond the immediate domain of medical care and placing substantial strain on other critical public institutions. For example, the COVID-19 pandemic, HIV/AIDS epidemic, and Ebola outbreaks have not only overwhelmed health care systems but also generated ripple effects that strained social service agencies and disrupted public education systems. The opioid epidemic is one of the most widespread and devastating public health crises, which has claimed the lives of nearly 727,000 people in the United States since 1999 and continues to cause significant social and institutional strain (CDC, 2023). The opioid overdose death rate rose sharply from 2.9 per 100,000 people in 1999 to 32.4 per 100,000 in 2021 (NCHS, 2023). The unprecedented rise in the opioid overdose deaths prompted the Center for Disease Control and Prevention (CDC) to declare this the worst drug overdose crisis in U.S. history (Kolodny et al., 2015).

This paper examines the impacts of the opioid epidemic on child maltreatment and how Child Protective Services (CPS) responded to this crisis. The opioid epidemic has profoundly destabilized families, particularly through its impacts on parents. Substance use significantly impairs parenting abilities, exposing their children to a higher risk of maltreatment (Wells, 2009; Rutherford and Mayes, 2017). Child maltreatment is associated with a host of long-lasting consequences including lower educational attainment, reduced earnings, lower employment, fewer assets, and higher involvement in crime later in life (Currie and Spatz Widom, 2010; Currie and Tekin, 2012; Cicchetti and Handley, 2019). These outcomes represent substantial losses in human capital and productivity.

In periods of heightened maltreatment risk, CPS plays a critical role in ensuring children’s safety by investigating maltreatment allegations and placing endangered children in foster care, a state-supervised living arrangement that provides temporary care outside the home. However, during the opioid epidemic, child maltreatment allegations have increased (Evans, Harris and Kessler, 2022), while foster care placement rates have remained flat (Gihleb, Giuntella and Zhang, 2022). Reconciling these seemingly inconsistent results in the literature requires a unifying framework for understanding CPS responses under heightened risk.

I exploit the reformulation of OxyContin, a nationwide supply-side drug policy which has received extensive attention from both academics and the media due to its unintended consequences. This supply disruption, which aimed to reduce OxyContin abuse by introducing an abuse-deterrent version of the original formulation, inadvertently exacerbated the opioid epidemic by driving users toward illicit, more addictive substances, resulting in higher rates of opioid overdoses, heroin-related arrests, and homicides (Powell and Pacula,

2021; Alpert, Powell and Pacula, 2018; Evans, Lieber and Power, 2019; Park, 2022; Tan, 2024). I begin by constructing a panel data using administrative data from the National Child Abuse and Neglect Data System (NCANDS) and the Adoption and Foster Care Analysis and Reporting System (AFCARS). The NCANDS Child Files provide report-level data on child abuse and neglect cases reported to CPS and include child identifiers that allow tracking of re-reports following initial investigations. I aggregate these data to the county and state levels and merge them with measures of prescription opioids and OxyContin misuse from the CDC and Alpert, Powell and Pacula (2018).

The identification strategy follows an exposure design, akin to Bartik or shift-share approaches, and exploits variation in states' and counties' differential exposure to the reformulation of OxyContin. Regions with higher rates of OxyContin misuse and greater prevalence of high-risk prescription opioids prior to the reformulation were more affected by the intervention. This is supported by the first stage results showing that areas with higher pre-reformulation OxyContin misuse rates and Schedule II prescription opioids per capita experienced greater declines in OxyContin misuse rates following the intervention, suggesting that the supply disruption had more "bite" in these regions. I use event study and difference-in-differences frameworks to compare outcome trends between regions with higher and lower exposure to the reformulation, estimating policy-relevant intent-to-treat (ITT) effect of the reformulation.

I find that despite heightened risk driven by the opioid epidemic, institutional responses did not adjust accordingly, leaving more children in potentially harmful home environments without adequate protection. The opioid epidemic led to a rise in maltreatment allegations in states and counties more exposed to the reformulation of OxyContin, relative to less affected regions. At the same time, it also led to a surge in the instances where CPS left children at home, but they were subsequently re-reported for maltreatment. Because the policy mandate clearly states that children should be placed in foster care when they are at risk of maltreatment at home, these findings point to a rise in false negatives, cases in which CPS did not place endangered children in foster care. Foster care placement rates remained unchanged and federal child welfare agency expenditures per allegation declined, pointing to a limited systemic response.

An increase in maltreatment allegations were primarily driven by reports from mandated reporters including educational, medical, legal, and social services personnel. A one standard deviation increase in exposure to the reformulation of OxyContin led to a 6% and 11% increase in maltreatment allegations immediately following the reformulation at the state and county levels, respectively. In the subsequent years, these effects grew to 14% and

23%, respectively. These findings suggest that the reformulation of OxyContin heightened maltreatment risk by affecting parents through various unintended consequences documented in the literature. Specifically, the rise in opioid overdoses, heroin-related fatalities and crimes associated with procuring illicit drugs, including heroin and synthetic opioids, likely contributed to increased maltreatment risk. The rise in maltreatment allegations was more pronounced for White children, consistent with existing literature documenting that White populations were disproportionately affected by the opioid epidemic compared to Black populations.

To measure false negatives, I use reinvestigations following the initial investigation as a primary measure of subsequent maltreatment at home. Reinvestigation is a widely used proxy for subsequent maltreatment in the child welfare literature and serves as a quality metric among policymakers (Antle et al., 2009; Putnam-Hornstein et al., 2015; Putnam-Hornstein, Prindle and Hammond, 2021; Baron et al., 2024a,b). I show that the results are robust to a range of alternative measures of subsequent maltreatment at home, including subsequent reinvestigations, CPS decisions that there is sufficient evidence to support the allegation (substantiations), and foster care placements following the initial investigation.

An increase in the false negatives indicate that a significant number of children exposed to heightened maltreatment risk during the opioid epidemic were not protected by the system. I further show that these results were not simply a mechanical consequence of a larger volume of allegations induced by the opioid epidemic. Instead, false negatives increased conditional on the number of allegations, that is, per 1,000 allegations. A one standard increase in exposure to the reformulation yields a 5% and 4% increase in false negatives per 1,000 allegations at the state and county levels, respectively. In the subsequent years, the estimated effects were 8% and 10%, respectively. These results suggest that a growing share of at-risk children systematically went undetected by CPS during this period.

Changes in foster care placements per 1,000 allegations reflect how CPS has adjusted its placement practices in response to the shock. If foster care placement rates did not increase despite rising risk, more children exposed to maltreatment risk would have remained at home without protection. I find that foster care placements per 1,000 allegations remained statistically unchanged following the reformulation of OxyContin. This suggests that CPS maintained a relatively fixed placement rate, rather than adjusting placement rates in response to changes in underlying risk.

In addition, I provide suggestive evidence that the misalignment between the increased

maltreatment risk and CPS placement decisions was not primarily driven by a shortage of foster homes. I show this by analyzing changes in the rate at which children were placed into congregate care, a setting where multiple children reside together and is generally considered a last resort. The congregate placement rate remained stable before and after the reformulation, providing suggestive evidence that the limited response to heightened risk was not primarily due to the shortage of foster homes.

The effectiveness and timeliness of CPS responses are closely tied to the level of expenditures by child welfare agencies. Insufficient fiscal responses to protect vulnerable children during the opioid epidemic have been raised by the media and widely discussed in public discourse. Funding for child welfare agencies comes from a mix of federal, state, and local sources. I find that child welfare agency expenditures from federal sources declined by 8% following the reformulation. This decline is consistent with the effects of sequestration under the 2011 Budget Control Act, which imposed automatic across-the-board cuts after Congress failed to enact a deficit-reduction plan. Taken together with the surge in allegations and false negatives, these findings suggest that the systemic response did not fully adjust to the heightened risks induced by the opioid epidemic.

This paper makes several contributions to the literature. First, it expands the existing research on the consequences of the opioid epidemic. Studies have documented that the opioid epidemic has led to higher health care costs ([White et al., 2005](#); [Leslie et al., 2019](#)), lower labor force participation rates and higher unemployment rates ([Harris et al., 2020](#)), higher crime rates ([Sim, 2023](#)), and more suicides ([Borgschulte, Corredor-Waldron and Marshall, 2018](#)). Regarding the impact on child welfare, there have been seemingly incongruous findings. [Evans, Harris and Kessler \(2022\)](#) documented the rise in maltreatment allegations whereas [Gihleb, Giuntella and Zhang \(2022\)](#) found stagnant or declining foster care placements. The absence of a unifying framework has made it difficult to reconcile these findings. This paper addresses this gap by providing empirical evidence that more at-risk children were left at home during the opioid epidemic, as CPS responses did not fully adjust to the rising severity of maltreatment risk. Furthermore, by examining the impact of the opioid epidemic on child welfare, this paper highlights the potential for the opioid epidemic to generate intergenerational consequences, contributing to long-term declines in mobility and widening socioeconomic inequality.

Second, this paper complements the literature examining supply-side drug policies. The economic literature has studied various policies aimed at curtailing the supply of abusable drugs. These policies include Prescription Drug Monitoring Programs ([Buchmueller and Carey, 2018](#); [Grecu, Dave and Saffer, 2019](#); [Gihleb, Giuntella and Zhang, 2022](#)), crackdowns

on doctors and pain clinic suppliers (Dobkin and Nicosia, 2009; Meinhofer, 2016; Soliman, 2023), triplicate prescription programs (Sigler et al., 1984; Weintraub et al., 1991; Hartzema et al., 1992; Simoni-Wastila et al., 2004), implementation of over-the-counter regulations (Dobkin, Nicosia and Weinberg, 2014), and reformulation of OxyContin (Alpert, Powell and Pacula, 2018; Evans, Lieber and Power, 2019; Evans, Harris and Kessler, 2022). While previous studies typically examine the impact of each supply-side drug policy in isolation, this paper explores a unique setting where an initial policy is followed by a government response, which may either mitigate or amplify the effects of the initial policy. This highlights the importance of considering policy complementarities and institutional spillovers, where the effectiveness or harm of one intervention depends on the response capacity of other systems (Coe and Snower, 1997; Orszag, 1998; Chang, Kaltani and Loayza, 2009).

Lastly, this paper adds to the growing body of economic literature on child welfare by examining the responsiveness of CPS foster care placement policies in the face of heightened maltreatment risk. While prior research has primarily focused on the causal effects of foster care placements on children’s long-term outcomes (Doyle Jr, 2007, 2008; Bald et al., 2022a; Baron and Gross, 2022; Gross and Baron, 2022), little attention has been paid to how CPS has responded to nationwide shocks that heightened maltreatment risk. This gap in the literature is particularly policy-relevant, as large-scale shocks to child maltreatment risk, such as drug epidemics (Evans, Harris and Kessler, 2022), economic downturns (Brooks-Gunn, Schneider and Waldfogel, 2013), and climate change (Evans, Gazze and Schaller, 2023), are not rare. By addressing this gap, the study underscores the need for adaptable child welfare policies capable of protecting vulnerable children during periods of crisis.

The remainder of this paper proceeds as follows. Section 2 provides background information that helps interpret the empirical results. Section 3 describes the data, the construction of supercounties, and presents descriptive statistics. Section 4 introduces the framework for defining false negatives and discusses its relevance to institutional responses to rising child maltreatment risk. Section 5 outlines the empirical strategy. Section 6 presents the main empirical results. Section 7 explores potential mechanisms for the main empirical results. Section 8 concludes.

## 2 Background

This section provides background information to contextualize the results presented in section 6. Section 2.A reviews empirical findings on the association between parental substance abuse and child maltreatment. Section 2.B offers an overview of the foster

care system in the U.S. Section 2.C provides documented facts about the opioid epidemic. Section 2.D discusses how the reformulation of OxyContin intensified the opioid epidemic and presumably increased child maltreatment risk.

## 2.A Parental Substance Abuse and Child Maltreatment

Existing literature has documented a strong association between parental substance abuse and child maltreatment. Drug addiction disrupts the neural circuits responsible for reward, stress reactivity, and regulation, which significantly impairs parenting abilities. This disruption lowers the salience of infant signals and increases the stress associated with caregiving (Wells, 2009; Rutherford and Mayes, 2017). Beyond impairing the capacity to care for children, substance abuse is also linked to outcomes that expose children to unsafe and violent environments (Walsh, MacMillan and Jamieson, 2003; White and Widom, 2008; Conners-Burrow, Johnson and Whiteside-Mansell, 2009; Raitasalo and Holmila, 2017). For example, seeking illicit drugs often necessitates engaging in criminal activities, further increasing the risk to children’s safety and well-being (Powis et al., 2000).

Since 2005, parental substance abuse has consistently been one of the most prevalent risk factors associated with child removal from home, following domestic violence, and accounting for 20% to 36% of all cases. In 2022, 23.8% of child maltreatment victims had the drug abuse caregiver risk factor (Children’s Bureau, 2022). Moreover, nearly half of mothers in maltreatment referrals, even without substance-related allegations, have a history of substance abuse, with opioids being the most common substance (Font and Goldstein, 2024). Parental substance abuse as a primary driver of child maltreatment suggests that the risk of maltreatment has risen significantly since the opioid epidemic, the worst drug overdose crisis in U.S. history.

## 2.B Foster Care in the United States

By the age of 18, approximately 6% of all children in the United States, including 16% of Native American children and 12% of Black children, will have experienced foster care at some point in their lives (Wildeman and Emanuel, 2014). When child maltreatment is suspected, any individual, including certain professionals who are mandated to report by law, can make a report by calling a hotline. These reports are then screened to determine if they warrant further investigation, with some being routed to local CPS offices. Once a report reaches CPS, investigators assess the evidence. If sufficient evidence supports the maltreatment allegation, the case is substantiated.



For substantiated cases, CPS has the authority to intervene based on the assessment of the risk. The most significant intervention is the removal of children from their homes and placement into foster care. The primary justification for placing a child in foster care is the potential risk of maltreatment if the child remains at home.<sup>1</sup> This means that CPS is responsible for predicting whether a child would be maltreated if left at home rather than placed in foster care. If CPS concludes that the reported children would be maltreated at home, they are placed in foster homes, where they reside while efforts are made to address the issues within their families. Parents are typically required to comply with a reunification plan, which may involve participation in rehabilitation or detox programs, especially when parental substance abuse is the cause of the child’s removal. The ultimate goal of CPS is to reunify families whenever it is safe and feasible, though this process often involves navigating complex and challenging dynamics between child welfare, parental rights, and the best interests of the child.

## 2.C Opioid Epidemic

From 1999 to 2022, nearly 727,000 people died from an opioid overdose in the U.S. (CDC, 2023). The opioid overdose death rate increased significantly from 2.9 per 100,000 people in 1999 to 32.4 per 100,000 in 2021 (NCHS, 2023). The unprecedented surge in deaths from opioid overdoses has led the Centers for Disease Control and Prevention (CDC) to declare this the worst drug overdose crisis in U.S. history (Kolodny et al., 2015).

Starting in the 1990s, changing perceptions of the opioids and updated treatment protocols led physicians to take a more aggressive approach in managing pain with opioids (Jones et al., 2018). The American Pain Society initiated a significant campaign to recognize pain as the fifth vital sign in 1995. Consequently, the Joint Commission on Accreditation of Healthcare Organizations (JCAHO) updated its guidelines in 2001 to mandate that physicians evaluate pain together with other vital signs during patient consultations (Phillips, 2000).

Moreover, Purdue Pharma introduced OxyContin in 1996, a prescription opioid painkiller that rapidly emerged as one of the top substances misused in the United States with global sales reaching \$35 billion (Cicero, Inciardi and Muñoz, 2005). OxyContin, which is the brand name for the opioid oxycodone hydrochloride, is a pain reliever with a controlled-release mechanism, intended to be taken orally without breaking or crushing. It was intended to be prescribed for the alleviation of moderate to severe pain stemming from conditions such as injuries, bursitis, neuralgia, arthritis, and cancer. Nevertheless, individuals could unlock

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<sup>1</sup>CPS policy manuals in many states explicitly mandate this as a core objective (MDHHS, 2020; NCDHHS, 2024).



the high dosage of oxycodone instantly by dissolving or crushing the pill, resulting in an immediate euphoric effect. Because of its significant potential for abuse, it is categorized as a Schedule II controlled substance. Recent economic studies demonstrated that the introduction and marketing of OxyContin account for a significant portion of overdose deaths, declines in the quality of life, and deteriorating children’s health over the past two decades, thereby suggesting it as a primary cause of the opioid epidemic ([Alpert et al., 2022](#); [Arteaga and Barone, 2022](#)).

## 2.D Reformulation of OxyContin and Child Welfare

In response to the escalating misuse of OxyContin, Purdue Pharma released an abuse-deterrent formulation of the drug, which was approved by the Food and Drug Administration (FDA) in 2010. The reformulated OxyContin was designed to be more difficult to dissolve or crush, thereby making it more challenging to abuse through ingestion, inhalation, or injection for immediate euphoric effects. Following the reformulation, there was a swift decrease in both the misuse of OxyContin and the distribution of oxycodone. Between 2010 and 2014, the national rate of self-reported OxyContin misuse dropped approximately by 40 percent, and for the first time, the overall legal distribution of oxycodone, as tracked by the DEA, saw a decline after the reformulation, ending a consistent rise that had been ongoing since 2000 ([Alpert, Powell and Pacula, 2018](#)).

However, studies have documented unintended consequences of the OxyContin reformulation. For example, several medical studies have found that the use of heroin and synthetic opioid, which are more potent than OxyContin, surged following the reformulation ([Coplan et al., 2013](#); [Cicero and Ellis, 2015](#); [Larochelle et al., 2015](#)). Moreover, economic studies identified a causal relationship between the OxyContin reformulation and an increase in opioid overdose deaths, total overdose deaths ([Powell and Pacula, 2021](#)), deaths related to heroin and synthetic opioids ([Alpert, Powell and Pacula, 2018](#); [Evans, Lieber and Power, 2019](#)), cases of Hepatitis and HIV ([Beheshti, 2019](#)), and homicides ([Park, 2022](#); [Tan, 2024](#)).

These results suggest that the reformulation of OxyContin increased children’s exposure to maltreatment risk at home through multiple channels. First, an increase in the opioid overdose deaths as well as total overdose deaths indicate that more parents were adversely affected by the opioid epidemic. Second, the neurobiological effects of heroin on parents would have heightened the risk. Compared to prescription opioids, heroin is more addictive ([Health-Americas, 2023](#)) and associated with more severe withdrawal symptoms ([Monico and Mitchell, 2018](#)), which likely impaired the parenting behaviors of users, increasing children’s exposure to maltreatment risk. Third, crimes associated with obtaining heroin may have

further elevated risks to children. Unlike prescription opioids, heroin is classified as a Schedule I substance under the Controlled Substances Act, making it illegal to purchase and often driving high-risk criminal behaviors to obtain it. [Mallatt \(2022\)](#) and [Powell and Pacula \(2021\)](#) documented an increase in heroin-related arrests following the reformulation, while [Park \(2022\)](#) and [Tan \(2024\)](#) documented a rise in homicide rates, suggesting that expanded illicit market activities led to more crimes. These findings indicate that parents who turned to heroin were more likely engaged in criminal activities, further exposing their children to risk at home.

### 3 Data

This section describes the data used in the empirical analyses. Section [3.A](#) provides details on the child welfare data used in the main analyses. Section [3.B](#) explains the construction of super counties to minimize censoring bias inherent in the child welfare data. Section [3.C](#) presents details on the opioid data used to construct the exposure measures. Section [3.E](#) reports descriptive statistics of the data.

#### 3.A Child Welfare Data

The primary data for child welfare are collected from the National Child Abuse and Neglect Data System (NCANDS) and the Adoption and Foster Care Analysis and Reporting System (AFCARS). NCANDS is a federally funded initiative that gathers yearly data on child abuse and neglect cases reported to CPS across the United States. Reporting to NCANDS by states is voluntary, yet the majority of states and the District of Columbia provide data during the timeframe of the analysis in this paper.<sup>2</sup> AFCARS, a data collection system required by federal mandate, gathers detailed information on all children protected under Title IV-B/E of the Social Security Act (Section 427). This database compiles data on every child in foster care and those adopted through the jurisdiction of state child welfare agencies. Since 1998, it has been compulsory for states to participate in this program. I use NCANDS Child Files 2004-2018 and AFCARS Foster Care Files 2004-2018 for the analysis.<sup>3</sup> For the analysis of foster care placements, I exclude six states that did not report placement outcomes in the

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<sup>2</sup>In the NCANDS data, Oregon and North Dakota were excluded because they did not report during the sample period.

<sup>3</sup>The analysis period of this paper covers 2004-2016. I use the report year rather than the submission year of each allegation because the former represents when the agency was notified of the case. Since report years and submission years do not always align, I include some waves of the NCANDS Child Files outside of the analysis period to minimize the loss of reported allegations in the sample.

NCANDS files.<sup>4</sup> For the analysis of false negatives, I exclude states where the successful matching rate between Child Files from  $t$  to  $t + 1$  is below 90% for more than two years within the period  $t \in \{2004, \dots, 2015\}$ .<sup>5</sup> Additionally, I exclude states that do not report foster care placements. These criteria yield a sample of 38 states and Washington D.C. for the analysis of false negatives.

### 3.B Construction of Super Counties

One feature of the NCANDS Child Files is that county identifiers are masked for counties with fewer than 700 allegations. Excluding all masked counties from the analysis sample would result in censoring bias. To minimize bias from this censoring, I construct “super counties” within each state using a two-step procedure. First, I identify counties that appear in every year of the Child Files. These are counties with more than 700 allegations annually throughout the sample period. There are 597 such “identified” counties in the data, accounting for approximately 69% of the U.S. population. Second, I create “super counties” in each state by aggregating outcome and control variables across all counties with fewer than 700 allegations. This step results in 46 super counties. The final balanced panel for the county-level analysis consists of 597 identified counties and the constructed super counties.

### 3.C OxyContin and Prescription Opioid Data

The identification strategy in this paper compares outcomes across regions that were more versus less exposed to the OxyContin reformulation. Ideally, exposure would be measured using pre-reformulation OxyContin misuse rates, following the approach in [Alpert, Powell and Pacula \(2018\)](#), since regions with higher misuse rates experienced larger declines in the misuse rates following the reformulation. The underlying idea is that the supply disruption had a greater impact in areas where OxyContin misuse was more prevalent. However, a practical limitation is that data on OxyContin misuse is only available at the state level, offering limited identifying variation. To address this, I complement the state-level analysis by leveraging county-level variation in Schedule II prescription opioids per capita, which is available at the county level, following the approach in [Evans, Harris and Kessler \(2022\)](#). Schedule II opioids, which include oxycodone, are highly addictive and subject to the most stringent federal regulations.

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<sup>4</sup>These states are Alabama, Georgia, Michigan, North Carolina, New York, and Pennsylvania

<sup>5</sup>The proportion of successful links of Child IDs across annual child files is based on the child’s date of birth and sex. These rates are reported in the file “Linking the NCANDS Child File Year to Year” provided by NDACAN.

For the state-level analysis, I utilize the measures for nonmedical OxyContin, covering the period from 2004 to 2009. I retrieve these measures from [Alpert, Powell and Pacula \(2018\)](#), which are based on data from the National Survey on Drug Use and Health (NSDUH). The NSDUH is a household survey that represents the national population and includes individuals aged 12 and above. It is the largest annual survey that collects data on substance use in the U.S. and specifically mentions OxyContin, distinguishing its nonmedical use. This data is only available at the state level, not at the county level. The pre-reformulation exposure to nonmedical OxyContin use is defined as the population-weighted rate in each state, combining survey data from 2004-2005 to 2008-2009. [Alpert, Powell and Pacula \(2018\)](#) showed that state-level OxyContin misuse rates from NSDUH align with both the legal supply data from the Automation of Reports and Consolidated Orders System (ARCOS) and opioid prescription data from the geocoded Medical Expenditure Panel Survey (MEPS).

For the county-level analysis, I use Schedule II prescription opioid data from the Centers for Disease Control and Prevention (CDC), covering the period from 2006 to 2009.<sup>6</sup> This data includes nearly 85% of all retail pharmacy providers in the United States, excluding hospitals. The pre-reformulation exposure measure is calculated as the population-weighted mean number of all Schedule II opioid prescriptions per capita in each county for the period 2006 to 2009.

### 3.D Child Welfare Expenditure Data

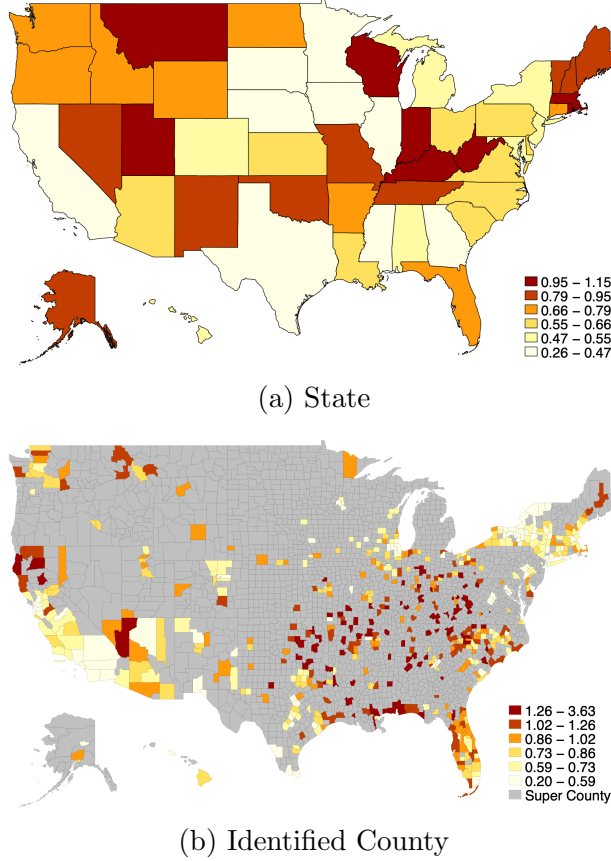
To analyze the fiscal response to the opioid epidemic, I use biannual, state-level data from the Child Welfare Financing Surveys for the years 2006 to 2018. These surveys were conducted by Child Trends and funded by the Annie E. Casey Foundation and Casey Family Programs, with financial data collected from all 50 states and the District of Columbia.

The surveys document state-reported child welfare expenditures across multiple funding sources, including federal, state, and local funds. Expenditures encompass a wide array of activities such as child protection services, case management, foster care, adoption services, administrative overhead, and staff salaries, as well as services for prevention and family preservation. The federal funds are drawn from both dedicated child welfare programs such as Titles IV-B and IV-E of the Social Security Act and non-dedicated programs including Medicaid, the Social Services Block Grant (SSBG), and Temporary Assistance for Needy Families (TANF).

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<sup>6</sup>While covering the period from 2004 would be ideal for consistency with the state-level analysis, which enables a more robust test of pretrends, the earliest available data from the CDC is from 2006. The same years of data were used in [Evans, Harris and Kessler \(2022\)](#)

Figure 1: Exposure Measure by State and Identified County



**Notes.** This figure illustrates the measure of exposure to the reformulation of OxyContin by state and identified counties. Figure (a) shows the rate of nonmedical use of OxyContin between 2004 and 2009 across states. Figure (b) shows Schedule II opioid prescriptions per capita across 597 identified counties between 2006 and 2009, where counties with less than 700 allegations in each state are classified as super counties.

Data collection was administered via mailed surveys to each state's child welfare administrator, as listed by the National Association of Public Child Welfare Administrators (NAPCWA). To ensure data quality, states were contacted for follow-up if reported spending in any category changed by more than 20% compared to the previous wave. After reconciliation, a one-page summary of reported expenditures was sent to each state for final review and confirmation. States were asked to verify their data or flag any necessary revisions, improving the reliability and consistency of reported figures across survey waves.

### 3.E Descriptive Statistics

Figure 1 shows the geographic distribution of the pre-reformulation OxyContin misuse rate by state and Schedule II prescription opioids per capita by identified county. The gray areas in Figure (b) represent super counties, where the exposure measure is computed by taking the population-weighted average of the number of Schedule II prescription opioids across masked counties. These figures illustrate significant variation in the exposure measure at both the state and county levels.

The main outcomes of interest are maltreatment allegations per 1,000 children, and false negatives measured both per 1,000 children and per 1,000 allegations. CPS follows clear guidelines, placing children in foster care only when there is a potential risk of maltreatment if they remain at home. Under this framework, CPS assesses the risk of subsequent maltreatment and predicts the likelihood of harm if a child is left at home. False negatives occur when CPS mistakenly leaves a child at home following an investigation, but the child is re-reported within six months. These are formally defined using the potential outcomes framework in section 4, and alternative proxies for subsequent maltreatment are used to test the robustness of the results in section 6.D.

False negatives measured per 1,000 children and per 1,000 allegations each offer distinct and complementary insights. False negatives per 1,000 children reflect how many at-risk children in the population were missed by the system, providing a direct measure of the broader welfare impact on the child population. This measure is relevant for estimating the overall impact of the opioid epidemic on child well-being at the population level. However, it speaks less directly to CPS responses, as CPS only observes and makes decisions about reported children, not the entire population. This is captured by false negatives per 1,000 allegations which reflect how CPS handled the cases it observed in response to the opioid epidemic.

Professional reporters include social services, medical, mental health, legal, education personnel, and child daycare providers. As shown in Table A3, allegations from professionals account for approximately 56% of all allegations, whereas allegations from non-professionals account for 20%. The rest of the allegations have unknown or anonymous reporters. I focus on allegations reported by professionals, as they are more likely to provide less biased estimates of underlying maltreatment risk. The reformulation of OxyContin was a nationwide intervention that influenced not only potential perpetrators but also potential reporters of maltreatment. This implies that reporters' abilities to identify and report maltreatment, as well as the consistency of reporting standards, may have been compromised by the intervention.

However, professionals are less likely to be affected in this way compared to non-professionals including neighbors, friends, and relatives, for two reasons. First, studies show significant variation in the rates of substance use disorders (SUD) across occupations, with professionals, as classified above, having the lowest rates from 2008 to 2012. (Bush and Lipari, 2016). Furthermore, these occupations demonstrated substantially lower rates of opioid-related overdose deaths, ranging from 4 to 15.4 deaths per 100,000 workers, compared to the occupational average of 25.1 during 2011-2015 in Massachusetts (Hawkins et al., 2019). The lower levels of substance dependence, particularly regarding opioids, suggest that professionals’ abilities to report maltreatment were likely less impaired following OxyContin’s reformulation, relative to non-professionals. Second, professionals are mandated reporters in most states and are provided with training and resources to support their ability to identify and report child maltreatment. This mandate should have enabled them to maintain greater consistency in reporting standards during periods of crisis compared to non-professionals.

Table 1: Summary Statistics

|                                    | Pre-reformulation |               |                 | Post-reformulation |               |                 |
|------------------------------------|-------------------|---------------|-----------------|--------------------|---------------|-----------------|
|                                    | Low-exposure      | High-exposure | <i>p</i> -value | Low-exposure       | High-exposure | <i>p</i> -value |
| <b>Panel A: States</b>             |                   |               |                 |                    |               |                 |
| Allegations per 1K Children        | 20.908            | 28.263        | 0.108           | 24.360             | 35.523        | 0.014           |
| False Negatives per 1K Children    | 2.060             | 2.639         | 0.083           | 2.529              | 4.065         | 0.010           |
| False Negatives per 1K Allegations | 53.752            | 59.177        | 0.200           | 60.626             | 74.360        | 0.017           |
| Percent Black                      | 14.526            | 9.582         | 0.054           | 14.724             | 10.061        | 0.079           |
| Percent Female                     | 50.876            | 50.826        | 0.803           | 50.826             | 50.759        | 0.727           |
| Percent Child                      | 27.900            | 26.720        | 0.094           | 26.501             | 25.342        | 0.174           |
| Unemployment Rate                  | 6.014             | 5.437         | 0.062           | 7.313              | 6.900         | 0.337           |
| Labor Force Participation Rate     | 66.077            | 65.603        | 0.640           | 63.630             | 62.959        | 0.499           |
| <b>Panel B: Counties</b>           |                   |               |                 |                    |               |                 |
| Allegations per 1K Children        | 22.865            | 28.629        | 0.054           | 24.190             | 35.401        | 0.002           |
| False Negatives per 1K Children    | 2.323             | 2.656         | 0.291           | 2.729              | 3.819         | 0.045           |
| False Negatives per 1K Allegations | 56.168            | 52.075        | 0.256           | 64.233             | 64.671        | 0.913           |
| Percent Black                      | 12.501            | 14.360        | 0.307           | 12.756             | 14.598        | 0.307           |
| Percent Female                     | 50.803            | 50.948        | 0.310           | 50.753             | 50.909        | 0.227           |
| Percent Child                      | 13.110            | 12.866        | 0.299           | 13.045             | 12.814        | 0.347           |
| Unemployment Rate                  | 5.891             | 6.644         | 0.029           | 7.079              | 7.576         | 0.134           |
| Labor Force Participation Rate     | 64.101            | 61.118        | 0.000           | 62.390             | 58.497        | 0.000           |

**Notes.** This table presents summary statistics for low- and high-exposure states and counties, separately for the pre-reformulation period (2004–2009 for states and 2006–2009 for counties) and the post-reformulation period (2010–2016). High-exposure states are those with an above-median pre-reformulation OxyContin misuse rate, while high-exposure counties are those with an above-median pre-reformulation prescription opioid supply per capita. The third and sixth columns report *p*-values from tests for the equality of means.

Table 1 presents summary statistics for low- and high-exposure states and counties, separately for the pre-reformulation period (2004–2009 for states and 2006–2009 for counties) and the post-reformulation period (2010–2016). High-exposure states are those with an



above-median pre-reformulation OxyContin misuse rate, while high-exposure counties are those with an above-median pre-reformulation Schedule II prescription opioid supply per capita. The third column reports  $p$ -values from tests for the equality of means. Prior to the reformulation, differences in allegations per 1,000 children and false negatives per 1,000 allegations between high- and low-exposure states were statistically insignificant but became significant in the post-reformulation period. Differences in false negatives per 1,000 children were significant in both periods, with a larger magnitude observed after the reformulation. Similar patterns are observed at the county level. Before the reformulation, differences in allegations per 1,000 children and false negatives per 1,000 children and per 1,000 allegations between high- and low-exposure counties were statistically insignificant. After the reformulation, differences in allegations per 1,000 children and false negatives per 1,000 children became statistically significant.

## 4 Conceptual Framework

In this section, I define false negatives in the context of CPS decision-making and discuss the implications of rising false negative rates for CPS responses during periods of heightened child maltreatment risk. A shock to maltreatment risk raises the proportion of children who would be maltreated if left in their homes. CPS’s response to this shock is reflected in its placement decisions for children who were exposed to maltreatment risk due to the causal effects of the shock, i.e., those who would not have been at risk in its absence. I evaluate CPS responses through prediction mistakes, as CPS placement decisions are made under clear policy guidelines. The primary justification for placing a child in foster care is the potential risk of maltreatment if the child remains at home.<sup>7</sup> This means that CPS is responsible for predicting whether a child would be maltreated if left at home rather than placed in foster care.

Each child has a potential for subsequent maltreatment  $Y_i^* \in \{0, 1\}$  where  $Y_i^* = 1$  indicates that a child would be maltreated if left at home following the CPS investigation. Let  $D_i \in \{0, 1\}$  denote the placement decision for child  $i$  where  $D_i = 1$  indicates that a child is placed in foster care. CPS’s response to a shock that exposes children to maltreatment is captured by their placement decisions for those whose  $Y_i^*$  switches from 0 to 1 as a result of the shock. A prediction mistake for these children is a false negative which occurs when an at-risk child

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<sup>7</sup>CPS policy manuals in many states explicitly mandate this as a core objective (MDHHS, 2020; NCDHHS, 2024).

$(Y_i^* = 1)$  is left at home ( $D_i = 0$ ). Formally, this is defined as:

$$FN_i = \mathbb{1}(Y_i^* = 1, D_i = 0)$$

The false negative rate can be expressed as follows:

$$\begin{aligned} \mathbb{P}(Y_i^* = 1, D_i = 0) &= \mathbb{P}(Y_i^* = 1) \times \mathbb{P}(D_i = 0 | Y_i^* = 1) \\ &= \underbrace{\mathbb{P}(Y_i^* = 1)}_{\text{Prevalence of at-risk children}} \times \left(1 - \underbrace{\mathbb{P}(D_i = 1 | Y_i^* = 1)}_{\text{Placement rate for at-risk children}}\right) \end{aligned}$$

where  $\mathbb{P}(Y_i^* = 1)$  represents the fraction of children at risk of subsequent maltreatment and  $\mathbb{P}(D_i = 1 | Y_i^* = 1)$  represents the placement rate for children with subsequent maltreatment potential. Any shock that affects the risk of maltreatment changes  $\mathbb{P}(Y_i^* = 1)$ . CPS response is captured by  $\mathbb{P}(D_i = 1 | Y_i^* = 1)$ , the placement rate for at-risk children. CPS can mitigate the rise in the false negative rate caused by an increase in at-risk children by raising the placement rate for these children. A rise in the false negative rate suggests that some of the children exposed to the risk by the shock were left in their homes.

The empirical challenge in measuring false negatives lies in the fact that  $Y_i^*$  is a latent variable. I use re-investigations initiated through the reports from professional reporters as a proxy for subsequent maltreatment. Details about the classification of professional reporters are explained in section 3. Re-investigations within six months of the initial investigation where the child was left at home are widely used as a proxy for subsequent maltreatment in both academic research and policy evaluation of child welfare systems (Antle et al., 2009; Putnam-Hornstein et al., 2015; Putnam-Hornstein, Prindle and Hammond, 2021; Baron et al., 2024a,b). Although this measure serves as an imperfect proxy for false negatives, re-investigations involve considerable interactions with authorities that entail a report to CPS and screening procedures in the central hotline center. In section 6.D, I show the robustness of the main results to alternative measures that involve decisions of CPS, including substantiation and foster care placement following the allegation by professionals.

## 5 Empirical Strategy

The empirical strategy is based on the exposure design, which is akin to Bartik or shift-share approaches. In this setting, the nationwide reformulation of OxyContin in 2010 serves as a common policy shock, while pre-reformulation OxyContin misuse rates (at the state level) or Schedule II opioid prescription rates (at the county level) serve as pre-determined measures

of exposure. Unlike the canonical Bartik design, the interaction of the exposure measure and the common shock is not used as an instrument for an endogenous treatment variable. Instead, the focus is on estimating the policy-relevant intent-to-treat (ITT) effect of the reformulation across areas with different levels of pre-reformulation exposure. Similar ITT approaches have been employed in prior studies of the OxyContin reformulation (Beheshti, 2019; Powell and Pacula, 2021; Evans, Harris and Kessler, 2022; Park, 2022).

I estimate event study and difference-in-differences regressions, leveraging the variation in states' and counties' exposure to the reformulation. For the event study, I estimate the following equations:

$$y_{st} = \sum_{k=2004}^{2016} \beta_k \mathbb{1}[t = k] \times \text{Exp}_s + \alpha_s + \gamma_t + X'_{st}\lambda + \epsilon_{st} \quad (1)$$

$$y_{ct} = \sum_{k=2006}^{2016} \gamma_k \mathbb{1}[t = k] \times \text{Exp}_c + \alpha_c + \gamma_t + X'_{ct}\lambda + u_{ct} \quad (2)$$

where  $y_{st}$  denotes the outcome in state  $s$  and year  $t$  and  $y_{ct}$  denotes the outcome in county  $c$  and year  $t$ .  $\alpha_s$  and  $\alpha_c$  denote state and county fixed effects, and  $\gamma_t$  denotes year fixed effects.  $\text{Exp}_s$  denotes a standardized pre-reformulation (2004-2010) rate of nonmedical OxyContin use and  $\text{Exp}_c$  denotes a standardized pre-reformulation (2006-2010) number of Schedule II opioid prescriptions per capita in county  $c$ .  $X_{st}$  denotes a vector of state- and time-varying covariates including the percentage of Black, female, and child population, unemployment rate, labor force participation rate and state- and time-varying policy indicators for a must-access Prescription Drug Monitoring Program, medical marijuana law and Medicaid expansion.  $X_{ct}$  denotes the same vector at the county level. I normalize the coefficient for year 2010 to zero and cluster standard errors at the state level. Regressions are estimated using weighted least squares, weighting by the child population in each state and year. The variables of interest are  $\beta_t$  and  $\gamma_t$  terms which identify how the outcomes in state  $s$  and county  $c$  in year  $t$  would have been different had its pre-reformulation OxyContin misuse rate or Schedule II opioid prescriptions per capita been one standard deviation higher.

I also estimate DID specifications to estimate the short-run and medium-run effects of

OxyContin’s reformulation. I estimate the following equations:

$$y_{st} = \beta_1 \times \text{Pre}_t^S \times \text{Exp}_s + \beta_2 \times \text{SRpost}_t \times \text{Exp}_s + \beta_3 \times \text{MRpost}_t \times \text{Exp}_s \quad (3)$$

$$+ \alpha_s + \gamma_t + X'_{st}\lambda + \epsilon_{st}$$

$$y_{ct} = \gamma_1 \times \text{Pre}_t^C \times \text{Exp}_c + \gamma_2 \times \text{SRpost}_t \times \text{Exp}_c + \gamma_3 \times \text{MRpost}_t \times \text{Exp}_c \quad (4)$$

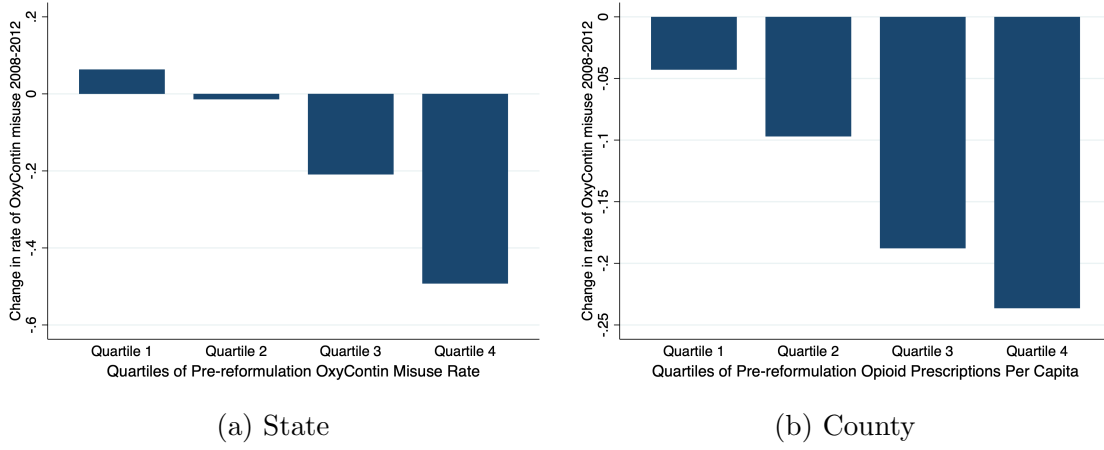
$$+ \alpha_c + \gamma_t + X'_{ct}\lambda + u_{ct}$$

where  $\text{Pre}_t^S$  and  $\text{Pre}_t^C$  take a value of 1 for years from 2004 to 2009 and 2006 to 2009, respectively.  $\text{SRpost}_t$  takes a value of 1 for years 2011 to 2013 and  $\text{MRpost}_t$  takes a value of 1 for years 2014 to 2016. Standard errors are clustered at the state level. Regressions are estimated using weighted least squares, weighting by the child population in each county and year. The variables of interest are  $\beta_1, \beta_2, \beta_3, \gamma_1, \gamma_2$  and  $\gamma_3$  terms which identify how the outcomes in state  $s$  and county  $c$  in year  $t$  would have been different in the pre-reformulation period, short-run and medium-run following the reformulation had its pre-reformulation OxyContin misuse rate or Schedule II opioid prescriptions per capita been one standard deviation higher. The effects are expected to be larger in the medium run, as existing literature on the consequences of the reformulation, including overdose deaths from opioids and heroin, as well as homicide rates, shows that these impacts tend to intensify over time. This is primarily because it takes time for the illicit drug market to evolve and for the resulting effects to manifest.

The identification strategy builds upon two assumptions: pre-reformulation OxyContin misuse rate and Schedule II opioid prescriptions per capita are (1) correlated with the “does” of the reformulation and (2) are not correlated with factors that could affect the differential trends in the outcomes following the reformulation, in the absence of the reformulation. Figure 2 presents the “first-stage” results, where Figure (a) plots the change in OxyContin misuse rates against quartiles of states’ pre-reformulation OxyContin misuse rates and Figure (b) plots the change in OxyContin misuse rates against quartiles of counties’ pre-reformulation Schedule II opioid prescriptions per capita. Both plots suggest that states and counties with higher exposure measures experienced larger declines in the OxyContin misuse rates following the reformulation. Regression results with covariates are reported in Table A1. These results are consistent with the findings of [Alpert, Powell and Pacula \(2018\)](#) and [Evans, Harris and Kessler \(2022\)](#), which show that the reformulation of OxyContin had a greater impact in states with higher levels of nonmedical OxyContin use and in counties with higher rates of Schedule II prescription opioid dispensing prior to the reformulation.

The second assumption requires that the pre-reformulation OxyContin misuse rates and

Figure 2: Exposure Measures and Change in OxyContin Misuse Rate between 2008-2012



**Notes.** This figure presents the change in the rate of OxyContin misuse between 2008 and 2012 across quartiles of the exposure measure at the state and county levels. Figure (a) plots the change in OxyContin misuse rates against quartiles of states' pre-reformulation OxyContin misuse rates. Figure (b) plots the change in OxyContin misuse rates against quartiles of counties' pre-reformulation Schedule II opioid prescriptions per capita. In both figures, OxyContin misuse rates are based on the measure from [Alpert, Powell and Pacula \(2018\)](#).

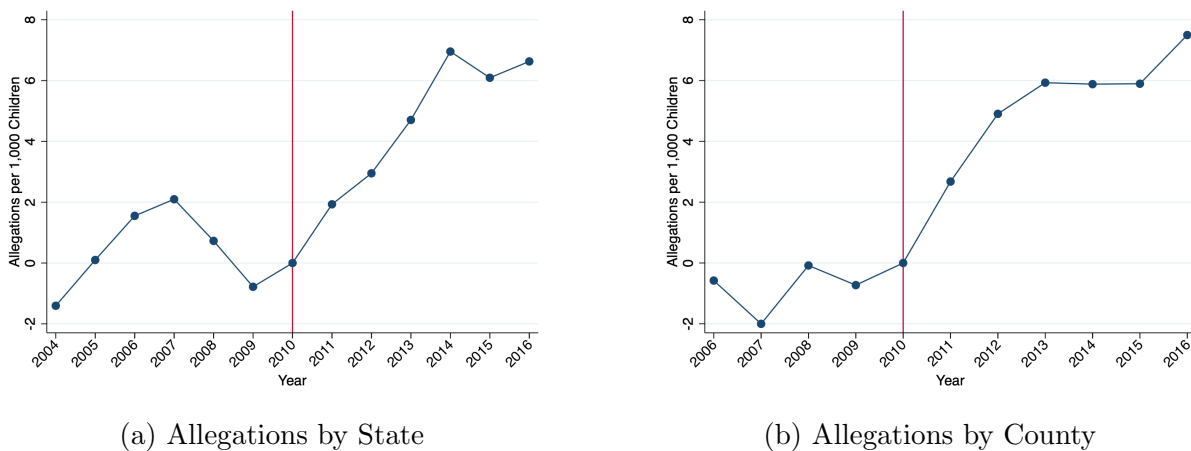
Schedule II prescription opioids per capita are exogenous to *changes*, as opposed to *levels* of the outcome variables ([Goldsmith-Pinkham, Sorkin and Swift, 2020](#)). Empirically, one can assess the plausibility of this assumption by performing pretrend tests as suggested in the recent literature on the exposure designs ([Goldsmith-Pinkham, Sorkin and Swift, 2020](#); [Borusyak, Hull and Jaravel, 2022](#)). In the next section, I show that pre-reformulation estimates of the outcomes are indistinguishable from zero in the event study and difference-in-differences results. I also test the balance of covariates by showing that the exposure measure is orthogonal to pre-reformulation changes in  $X_{st}$  and  $X_{ct}$ . These results are reported in Table [A2](#).

## 6 Results

In this section, I present the empirical results. Section [6.A](#) provides preliminary evidence by illustrating descriptive trends in the data. Section [6.B](#) presents the main causal results. Section [6.C](#) examines heterogeneous effects by characteristics of children and allegations. Section [6.D](#) assesses the robustness of the main findings.

## 6.A Preliminary Evidence

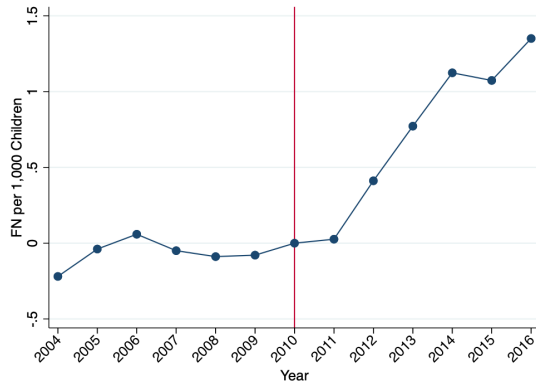
Figure 3: Differential Trends in Maltreatment Allegations per 1,000 Children



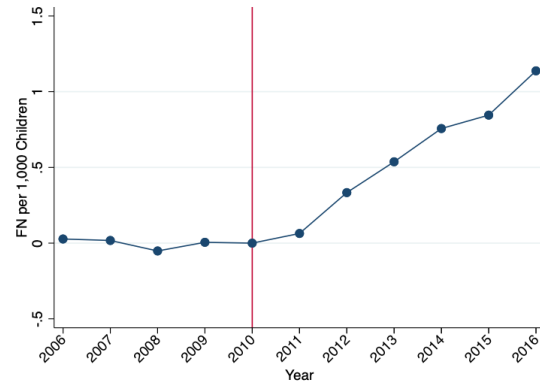
**Notes.** This figure illustrates the trends in the differences in maltreatment allegations per 1,000 children between high- and low-exposure states and counties. High-exposure states refer to states where the pre-reformulation (2004-2009) OxyContin misuse rates are above the median. High-exposure counties refer to counties where the pre-reformulation (2006-2009) Schedule II opioid prescriptions per capita are above the median. Differences in 2010 are normalized to 0.

Before presenting the main event study and difference-in-differences results, I examine trends in outcome differences between high- and low-exposure states and counties, normalizing the differences in 2010 to 0. High-exposure states are defined as those where pre-reformulation (2004–2009) OxyContin misuse rates are above the median, while high-exposure counties are those where pre-reformulation (2006–2009) Schedule II opioid prescriptions per capita are above the median. Figure 3 presents differential trends in maltreatment allegations per 1,000 children. Prior to the reformulation, the differences in maltreatment allegations relative to 2010 are bounded between -2 and 2 at the state level and between -2 and 0 at the county level. These differences increased substantially following the reformulation, suggesting that regions more heavily affected by the opioid supply disruption experienced relatively larger increases in maltreatment allegations.

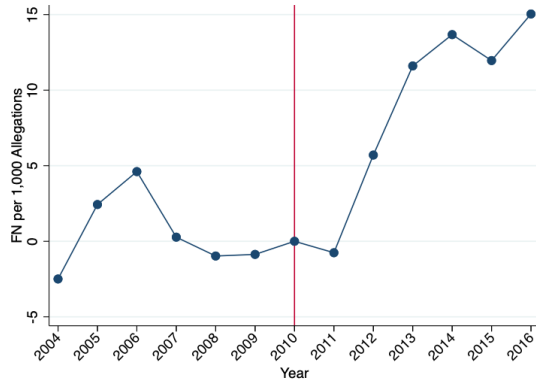
Figure 4: Differential Trends in False Negative Rates



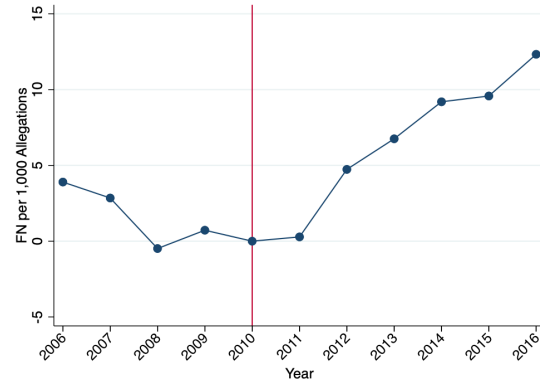
(a) FN by State, per 1,000 Children



(b) FN by County, per 1,000 Children



(c) FN by State, per 1,000 Allegations



(d) FN by County, per 1,000 Allegations

**Notes.** This figure illustrates the trends in the differences in false negatives per 1,000 children and 1,000 allegations between high- and low-exposure states and counties. False negatives refer to cases in which children were re-reported within six months after not being placed in foster care following the initial investigation. High-exposure states refer to states where the pre-reformulation (2004-2009) OxyContin misuse rates are above the median. High-exposure counties refer to counties where the pre-reformulation (2006-2009) Schedule II opioid prescriptions per capita are above the median. Differences in 2010 are normalized to 0.

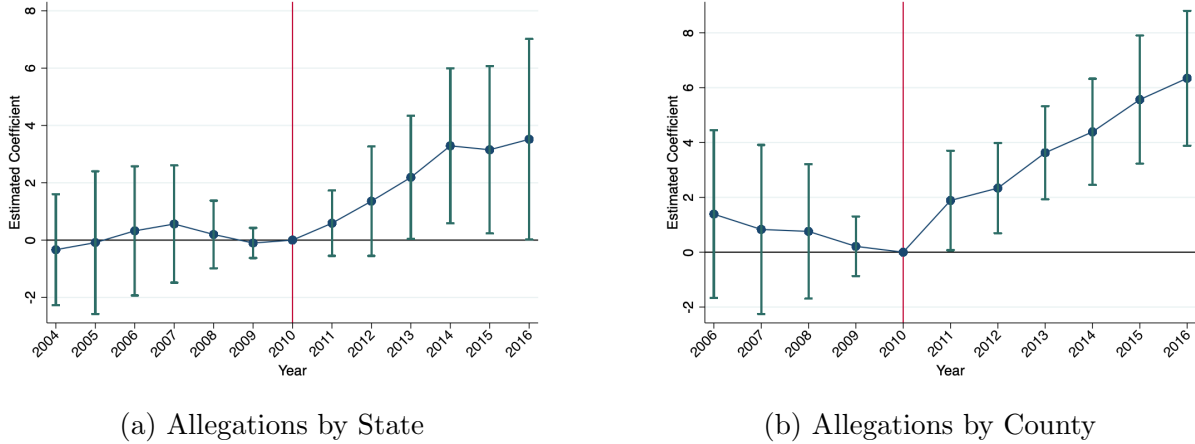
Figure 4 presents trends in false negatives per 1,000 children and 1,000 allegations. Prior to the reformulation, the differences in false negatives were close to 0 across both measures. Following the reformulation, these differences increased substantially, suggesting that relatively more at-risk children were left in their homes in high-exposure states and counties.



## 6.B Main Results

### 6.B.1 Maltreatment Allegations

Figure 5: Event Study Results for Allegations per 1,000 Children



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 and Equation 2 that are adjusted for within-state clustering. The dependent variable is maltreatment allegations reported by professionals per 1,000 children. Figure (a) is based on a state-level analysis, whereas Figure (b) is based on a county-level analysis. Regressions are weighted by child population.

Figure 5 and Table 2 present the event study and difference-in-differences results for allegations per 1,000 children. Allegations increased at both the state and county levels. At the state level, a one standard deviation increase in the pre-reformulation OxyContin misuse rate corresponds to a 5.7% increase in the allegations in the short run, followed by a 13.7% increase in the medium run. At the county level, a one standard deviation increase in pre-reformulation Schedule II prescription opioids per capita yields a 10.9% and 22.5% increase in the allegations in the short and medium run, respectively.

An increase in the allegations indicates that the underlying maltreatment risk increased following the reformulation of OxyContin. This heightened risk is presumably driven by the exacerbation of the opioid epidemic after the reformulation. As documented in the existing literature, the reformulation led to increases in opioid overdose deaths, use of heroin and synthetic opioids, cases of hepatitis and HIV, homicides, and heroin-related arrests. Given the strong link between parental substance use and child maltreatment, these findings reflect elevated risks faced by vulnerable children during periods of rising opioid misuse.

Table 2: Difference-in-Differences Results for Allegations per 1,000 Children

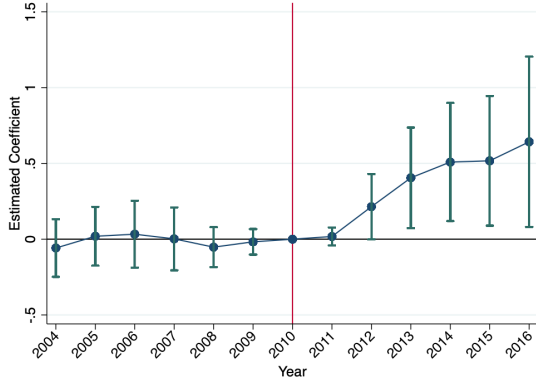
|                      | (1)                           | (2)                            |
|----------------------|-------------------------------|--------------------------------|
|                      | Allegations<br>State          | Allegations<br>County          |
| Pre-reformulation    | 0.100<br>(0.763)              | 0.790<br>(1.169)               |
| Short-run            | 1.366*<br>(0.786)<br>[5.7%]   | 2.612***<br>(0.811)<br>[10.9%] |
| Medium-run           | 3.288**<br>(1.486)<br>[13.7%] | 5.406***<br>(1.068)<br>[22.5%] |
| Mean (2010)          | 23.976                        | 23.976                         |
| R-Squared            | 0.869                         | 0.867                          |
| Observations         | 634                           | 7073                           |
| State or County FE   | Yes                           | Yes                            |
| Year FE              | Yes                           | Yes                            |
| $X_{st}$ or $X_{ct}$ | Yes                           | Yes                            |

**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where the dependent variable is maltreatment allegations per 1,000 children. Column (1) is based on a state-level analysis whereas Column (2) is based on a county-level analysis. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

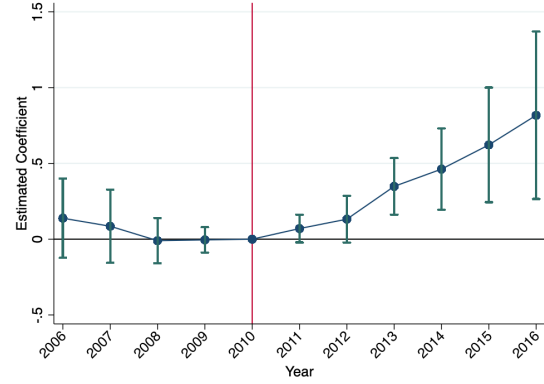
\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## 6.B.2 False Negative Rate

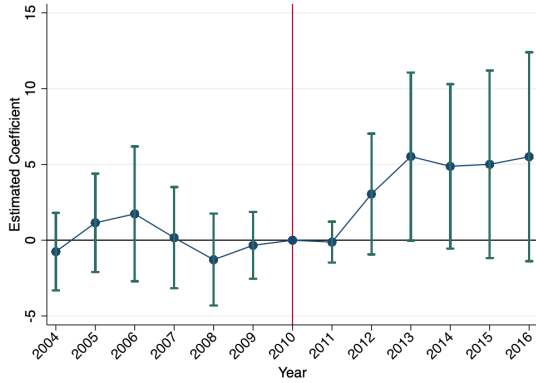
Figure 6: Event Study Results for False Negative Rates



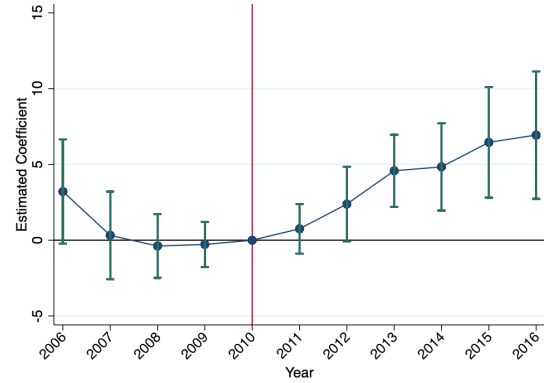
(a) FN by State, per 1,000 Children



(b) FN by County, per 1,000 Children



(c) FN by State, per 1,000 Allegations



(d) FN by County, per 1,000 Allegations

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 and Equation 2 that are adjusted for within-state clustering. The dependent variables are false negative rates. Figures (a) and (b) report the results for false negatives per 1,000 children whereas Figures (c) and (d) report the results for false negatives per 1,000 allegations. Figures (a) and (c) are based on a state-level analysis whereas Figures (b) and (d) are based on a county-level analysis. Regressions are weighted by child population.

Figure 6 and Table 3 present the results for false negative rates. These results suggest that the false negative rates significantly increased following the reformulation. A one standard deviation increase in the pre-reformulation OxyContin misuse rate yields a 14.5% and 37.9% increase in the false negatives per 1,000 children in the short and medium run, respectively. At the county level, a one standard deviation increase in the pre-reformulation Schedule II prescription opioids per capita yields a 12.7% and 43.5% increase in the short and medium

Table 3: Difference-in-Differences Results for False Negative Rates

|                      | (1)<br>FN<br>Children<br>State | (2)<br>FN<br>Children<br>County | (3)<br>FN<br>Allegations<br>State | (4)<br>FN<br>Allegations<br>County |
|----------------------|--------------------------------|---------------------------------|-----------------------------------|------------------------------------|
| Pre-reformulation    | -0.015<br>(0.072)              | 0.049<br>(0.082)                | 0.042<br>(1.309)                  | 0.658<br>(1.431)                   |
| Short-run            | 0.209**<br>(0.091)<br>[14.5%]  | 0.182***<br>(0.054)<br>[12.7%]  | 2.780*<br>(1.632)<br>[4.5%]       | 2.544**<br>(1.134)<br>[4.1%]       |
| Medium-run           | 0.545**<br>(0.221)<br>[37.9%]  | 0.626***<br>(0.197)<br>[43.5%]  | 4.979*<br>(2.939)<br>[8.1%]       | 5.989**<br>(2.779)<br>[9.7%]       |
| Mean (2010)          | 1.438                          | 1.438                           | 61.838                            | 61.838                             |
| R-Squared            | 0.729                          | 0.730                           | 0.789                             | 0.746                              |
| Observations         | 506                            | 4587                            | 506                               | 4587                               |
| State or County FE   | Yes                            | Yes                             | Yes                               | Yes                                |
| Year FE              | Yes                            | Yes                             | Yes                               | Yes                                |
| $X_{st}$ or $X_{ct}$ | Yes                            | Yes                             | Yes                               | Yes                                |

**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where the dependent variables are false negatives rates. Columns (1) and (2) report the results for false negatives per 1,000 children whereas Columns (3) and (4) report the results for false negatives per 1,000 allegations. Columns (1) and (3) are based on a state-level analysis whereas Columns (2) and (4) are based on a county-level analysis. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

run, respectively.

These findings suggest that more children in the population were exposed to maltreatment risk following the reformulation, as reflected by the rise in maltreatment allegations, and that an increasing number of at-risk children were left at home. More broadly, they indicate that the opioid epidemic systematically increased the number of children harmed without adequate institutional protection, highlighting its broad consequences for child welfare. While this is a striking result, it is important to acknowledge that some of the increase in false negatives may occur mechanically due to an increase in the allegations. By definition, a false negative requires that a child is at risk, reported to CPS, and subsequently left at home. As such, conditional on a fixed decision rule, an increase in the number of allegations

can lead to more false negatives simply due to a larger pool of reported children. Therefore, while the rise in false negatives per 1,000 children is a meaningful indicator of the opioid epidemic’s broader impact on child welfare at the population level, it may not fully reflect CPS responses.

However, an increase in the false negatives per 1,000 allegations suggests that such patterns are not solely driven by a larger pool of reported children. At the state level, a one standard deviation increase in the pre-reformulation OxyContin misuse rate yields a 4.5% and 8.1% increase in the false negatives per 1,000 allegations in the short and medium run, respectively. At the county level, a one standard deviation increase in pre-reformulation opioid prescriptions per capita yields a 4.1% and 9.7% increase in the short and medium run, respectively. A rise in the false negatives conditional on allegations indicate that CPS systemically left more at-risk children at home following the reformulation.

## 6.C Heterogeneity Analysis

Table A3 presents the percentage breakdown of allegations by maltreatment type, child characteristics, and report sources, separately for all allegations and for those made by professionals during the sample period from 2004 to 2016. Neglect is the most common type of maltreatment, comprising 60.4% of all allegations, followed by physical abuse, which accounts for 21.8%. Among professional reports, neglect and physical abuse remain the most frequent types of maltreatment, making up approximately 77% of these allegations. Education, legal, social services, and medical personnel are the most common sources of reports, accounting for 51% of all allegations. Among reports made by professionals, these groups represent 91% of the total.

Figures A5 and A6, along with Tables A6 and A7, present the results for allegations by child characteristics: race, gender, and age. The results indicate that the effects of the reformulation were heterogeneous across racial groups, with maltreatment allegations for White children increasing at a higher rate than for Black children. These findings are consistent with the existing literature, which documents that the opioid epidemic has disproportionately impacted White populations compared to Black populations. Specifically, White individuals are twice as likely as non-White minorities to use prescription opioid painkillers for non-medical purposes (Netherland and Hansen, 2016).

This racial disparity in opioid misuse is largely attributed to non-White minorities receiving inadequate pain management in various healthcare settings (Burgess et al., 2006; Sabin et al., 2009; Santoro and Santoro, 2018; Nicolette Harris, Long et al., 2021). Consequently, White

people, who are more frequently prescribed opioid pain relievers, were likely more affected by the disruption in opioid supply following the reformulation. This increased exposure to opioids among White people may have led to a higher rate of associated negative outcomes, including increased rates of child maltreatment allegations.

Figures [A1](#), [A2](#), [A3](#), and [A4](#), along with Tables [A4](#) and [A5](#), present the results for allegations categorized by maltreatment type and report source. Neglect and physical abuse, which together account for 82% of all allegations and 77% of professional allegations, both increased following the reformulation. In terms of report sources, the increase in allegations was primarily driven by education, social services, and medical personnel.

Figures [A7](#), [A9](#), [A8](#), [A10](#), [A12](#) and [A11](#), along with Tables [A9](#), [A8](#), [A11](#), and [A10](#) report the results for false negatives by maltreatment types and child characteristics. These results show that false negatives increased across all types of cases and demographics of children. The rise was more pronounced for White children than for Black children, measured per 1,000 children, suggesting that a greater number of at-risk White children were left at home unprotected following the reformulation. This pattern is largely driven by the disproportionate impact of the opioid epidemic on White children, increasing the the number of at-risk White children slipping through the system.

## 6.D Robustness Checks

Figures [A14](#), [A13](#) and Tables [A13](#), [A12](#) present the event study and difference-in-differences results for alternative measures of false negatives. These measures are based on CPS decisions following allegations made by professionals, specifically focusing on substantiation and foster care placements within six months of an initial investigation that resulted in the child being left at home. The robustness of these results suggests that increases in allegations and false negatives are not merely a result of disagreements between reporters and CPS regarding the maltreatment potential of children but rather reflect a misalignment between rising underlying risk and CPS placement decisions. Figures [A16](#), [A15](#) and Tables [A15](#), [A14](#) present the results based on the alternative time frames for defining false negatives. The results are robust to subsequent allegations within 3 months, 9 months, and 12 months following the initial investigation.

Figure [A17](#) and Table [A16](#) present the event study and difference-in-differences results for false negatives based on the provision of services. Children and families involved in child welfare investigations may receive post-reponse services between the report date and up to 90 days after the disposition date. Post-response services focus on ensuring the child's

safety and are typically guided by an evaluation of the family’s circumstances, including their needs for support and their strengths.<sup>8</sup> An alternative measure of false negatives would be to define it as a case where the reported child was not provided with post-response service but was reported within 6 months of the focal investigation. Figure A17 and Table A16 show that the false negative rate based on this measure increased following the reformulation, consistent with results based on the definition of false negatives using foster care placements as a treatment.

Alternative empirical specifications have been estimated for robustness checks. Figure A18 and Table A18 present results based on a binary treatment, where the exposure measure is defined as an indicator taking the value of one if the state’s pre-reformulation OxyContin misuse rate is above the median and if the county’s pre-reformulation opioid prescriptions per capita are above the median. Table A17 shows the regression results where the exposure measure is binned into quartiles, with the first quartile serving as the reference group. Finally, Figure A19 and Table A19 report findings excluding the year 2009 in the event study and difference-in-differences specifications. Since Purdue Pharma ceased shipping the old formulations in August 2010, part of 2010 may be considered a treated period. These results confirm that the main findings are robust to these alternative specifications.

## 7 Mechanisms

This section explores potential mechanisms behind the rise in false negatives following the reformulation of OxyContin. Section 7.A presents results on foster care placements, which reflect CPS placement tendencies. Section 7.B provides suggestive evidence on the role of supply of foster homes. Section 7.C reports results on child welfare agency expenditures, capturing the fiscal response to the opioid epidemic.

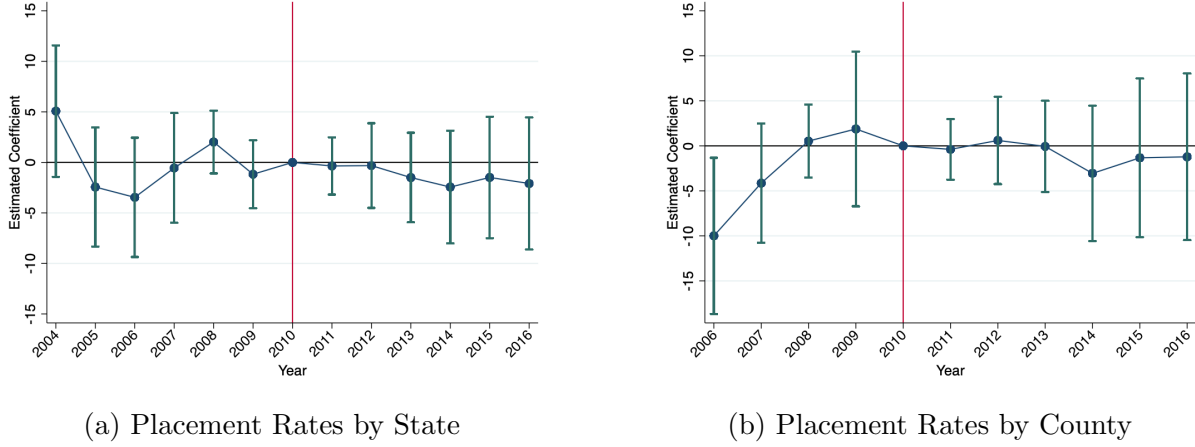
I find that changes in the foster care placement rates were insignificant. Using congregate placement rates as a proxy for the supply of foster homes, I find limited a limited impact of the opioid epidemic on the supply of foster homes. Finally, I document a decline in child welfare agency expenditures from federal funding sources, suggesting that a limited fiscal response may have contributed to the rise in false negatives.

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<sup>8</sup>Post-response services include the following: family support, family preservation, foster care, adoption, case management, counseling, day care, education and training, employment, family planning, health-related and home health, home-based, housing, independent and transitional living, information and referral, legal, mental health, pregnancy and parenting, respite care, special, substance abuse, and transportation services.

## 7.A Foster Care Placements

Figure 7: Event Study Results for Placement per 1,000 Allegations



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 and Equation 2 that are adjusted for within-state clustering. The dependent variables are foster care placements per 1,000 allegations. Figure (a) is based on a state-level analysis, whereas Figure (b) is based on a county-level analysis. Regressions are weighted by child population.

Foster care placements per 1,000 allegations capture CPS's placement tendencies. Flat placement rates despite rising severity of allegations are a potential mechanism underlying the increase in false negatives per 1,000 allegations. Figure 7 and Table 4 present the event study and difference-in-differences results for foster care placement rates. The results show that changes in foster care placements per 1,000 allegations were statistically indistinguishable from zero. Tables A20 and A21 show that these results hold across maltreatment types and demographics of the children. These results suggest that CPS did not adjust placement tendencies in response to the rising severity of allegations, resulting in more at-risk children being left at home. These findings provide suggestive evidence of rule-of-thumb placement norms, in which placement decisions follow a predetermined cap or target rate in each period rather than responding dynamically to changes in underlying risk.

## 7.B Foster Homes

A stagnant foster care placement rate in response to rising maltreatment risk may reflect a shortage of available foster homes. I provide suggestive evidence on the role of capacity constraints in shaping the response to the opioid epidemic. Foster homes are typically



Table 4: Difference-in-Differences Results for Placement Rates

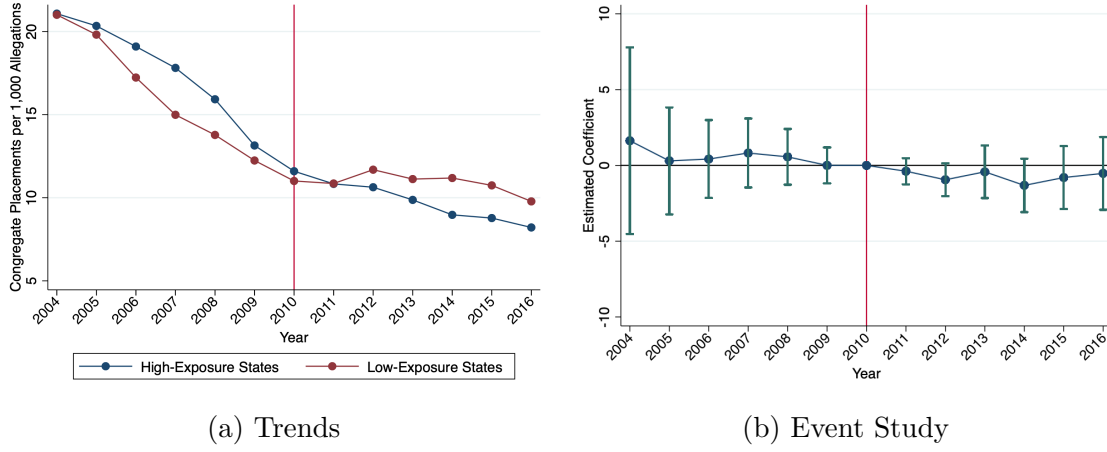
|                      | (1)<br>State<br>Placement    | (2)<br>County<br>Placement   |
|----------------------|------------------------------|------------------------------|
| Pre-reformulation    | -0.055<br>(1.673)            | -2.719<br>(2.758)            |
| Short-run            | -0.714<br>(1.674)<br>[-1.0%] | -0.033<br>(1.893)<br>[0.0%]  |
| Medium-run           | -1.906<br>(2.862)<br>[-2.7%] | -1.972<br>(4.229)<br>[-2.8%] |
| Mean (2010)          | 71.256                       | 71.256                       |
| R-Squared            | 0.748                        | 0.740                        |
| Observations         | 506                          | 4587                         |
| State or County FE   | Yes                          | Yes                          |
| Year FE              | Yes                          | Yes                          |
| $X_{st}$ or $X_{ct}$ | Yes                          | Yes                          |

**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where the dependent variables are foster care placements per 1,000 allegations. Column (1) is based on a state-level analysis whereas Column (2) is based on a county-level analysis. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

categorized as either kinship care or unrelated foster families. When no foster homes are available, children may be placed in congregate care, a structured, supervised setting where multiple children or youth live together rather than in a family-based environment. Congregate care includes group homes, which are licensed or approved facilities providing 24-hour care for generally 7 to 12 children, and institutions, larger facilities operated by public or private agencies that care for more than 12 children.

Figure 8: Congregate Placement Rate



**Notes.** Figure (a) illustrates the trends in congregate placements per 1,000 allegations from 2004 to 2016 separately for states with a pre-reformulation OxyContin misuse rate above or below and equal to the sample mean of 0.657. Figure (b) reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 that are adjusted for within-state clustering, where the dependent variable is congregate placements per 1,000 allegations.

Congregate care is generally considered a last resort for foster children (Bald et al., 2022b). Research has documented negative outcomes for children in congregate care and congregate placement rates have gradually declined (Lee and Thompson, 2008; Ryan et al., 2008; Robst, Armstrong and Dollard, 2011). Figure 8 shows these trends, with congregate placements per 1,000 allegations declining in both high- and low-exposure states. If foster home shortages were more severe in high-exposure states, congregate placement rates would likely have increased more or declined less in those states relative to low-exposure states.

For the analysis of capacity constraints, I use the same subsample of 38 states and Washington, D.C., as in the false negative analysis. This analysis is conducted at the state level only, as the AFCARS Foster Care Files mask data for counties with fewer than 700 foster care placements, resulting in most counties being excluded. Figure 8 and Table 5 presents the event study and difference-in-differences results for congregate placements rates. The results suggest that the congregate placement rate remained stable following the reformulation. This suggests that a foster home shortage has not been the primary driver of trends in foster care placements.

Table 5: Difference-in-Differences Results for Congregate Placement Rate

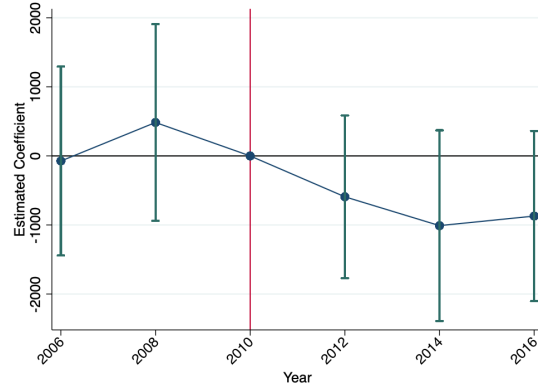
|                   | (1)                          |
|-------------------|------------------------------|
| Pre-reformulation | 0.594<br>(1.264)             |
| Short-run         | -0.576<br>(0.571)<br>[-5.1%] |
| Medium-run        | -0.864<br>(0.985)<br>[-7.7%] |
| Mean (2010)       | 11.221                       |
| R-Squared         | 0.860                        |
| Observations      | 506                          |
| State FE          | Yes                          |
| Year FE           | Yes                          |
| $X_{st}$          | Yes                          |

**Notes.** This table reports point estimates and standard errors from Equation 3, where the dependent variable is congregate placements per 1,000 allegations. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

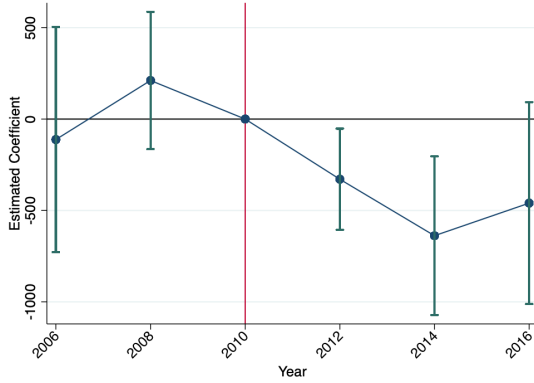
\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## 7.C Child Welfare Expenditures

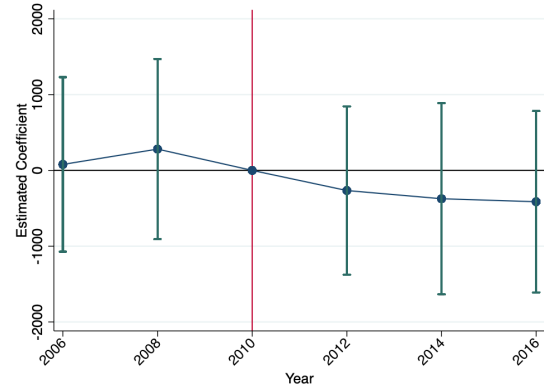
Figure 9: Event Study Results for Child Welfare Expenditures



(a) Total



(b) Federal



(c) State

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 that is adjusted for within-state clustering. The dependent variables are child welfare expenditures per allegation. Figure (a) is based on total expenditures, whereas Figures (b) and (c) are based on federal and state expenditures, respectively. Regressions are weighted by child population.

The capacity of CPS to respond effectively is closely tied to financial resources available for child protection. Child welfare agencies are funded through a combination of federal, state, and local sources. Federal funding includes both dedicated and non-dedicated streams. Dedicated funding primarily comes from Titles IV-B and IV-E of the Social Security Act, which support core child welfare activities. Non-dedicated federal sources, such as Medicaid, the Social Services Block Grant (SSBG), and Temporary Assistance for Needy Families

Table 6: Difference-in-Differences Results for Child Welfare Expenditures per Allegation

|                                | (1)                              | (2)                                | (3)                              |
|--------------------------------|----------------------------------|------------------------------------|----------------------------------|
|                                | Total                            | Federal                            | State                            |
| Post $\times$ Exp <sub>s</sub> | -946.325<br>(639.344)<br>[-6.7%] | -503.692**<br>(227.440)<br>[-7.8%] | -459.452<br>(583.072)<br>[-6.0%] |
| Mean                           | 14080                            | 6444.44                            | 7608.29                          |
| R-Squared                      | 0.833                            | 0.819                              | 0.804                            |
| Observations                   | 234                              | 234                                | 233                              |
| State FE                       | Yes                              | Yes                                | Yes                              |
| Year FE                        | Yes                              | Yes                                | Yes                              |
| $X_{st}$                       | Yes                              | Yes                                | Yes                              |

**Notes.** This table reports point estimates and standard errors from Equation 5, where the dependent variables are child welfare expenditures per allegation. Column (1) is based on total expenditures, whereas Columns (2) and (3) are based on federal and state expenditures, respectively. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in the pre-reformulation period (2006-2009) are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

(TANF), can also be used to support child welfare services, though they are allocated for broader purposes. In addition to federal funding, states contribute their own resources to operate federally mandated programs and to cover expenses not reimbursed by the federal government. Since 2008, state and local governments have accounted for approximately 55% to 57% of total child welfare expenditures, with the remaining share covered by federal sources.

To analyze the fiscal response to the opioid epidemic, I use biannual data from the Child Welfare Financing Survey and estimate event study and difference-in-differences models. For the difference-in-differences analysis, I estimate the following equation.

$$y_{st} = \beta \times \text{Post} \times \text{Exp}_s + \alpha_s + \gamma_t + X'_{st}\gamma + \epsilon_{st} \quad (5)$$

where  $t \in \{2006, 2008, 2010, 2012, 2014, 2016\}$  and Post is an indicator variable for years after 2010. Figure 9 and Table 6 show that expenditures of federal funds decreased by 484 dollars per allegation, representing an 8.7% decline from the pre-reformulation period.

The decline in federal expenditures per allegation following the reformulation aligns with broader fiscal pressures. Under the 2011 Budget Control Act, sequestration took effect in 2013 after Congress failed to enact a deficit-reduction plan, triggering automatic

across-the-board cuts. Flexible federal streams such as Title IV-B and the Social Services Block Grant were reduced, constraining the federal share of child welfare spending while maltreatment risk from the opioid epidemic was intensifying. As a result, agencies likely operated with fewer resources per case, a constraint that is consistent with the observed rise in false negatives.

## 8 Conclusion

The consequences of public health crises often extend beyond the health care system and place substantial strain on other critical public institutions. The opioid epidemic, widely regarded as the worst drug overdose crisis in U.S. history, has had far-reaching impacts and imposed substantial costs on society. Given the strong association between parental substance abuse and child maltreatment, understanding the effects of the opioid epidemic on child welfare and evaluating the responsiveness of CPS to mitigate these consequences are critical areas of academic and public policy research.

Leveraging cross-state variation in pre-reformulation OxyContin misuse rates and cross-county variation in pre-reformulation Schedule II prescription opioids per capita, I show that the opioid epidemic led to an increase in maltreatment allegations. Despite heightened risk, institutional responses did not adjust accordingly, leading to an increase in the recurrence of maltreatment among children who were not initially placed in foster care. Foster care placement rates remained unchanged and federal child welfare expenditures declined, pointing at limited systemic adjustment.

The findings of this paper uncover the impacts of the opioid epidemic on child welfare as well as its ripple effects on public institutions beyond the health care system. Given the long-term consequences of child maltreatment, including lower educational attainment, poorer health, and diminished economic mobility, these results suggest that the opioid epidemic can have long-lasting intergenerational impacts. More broadly, these findings underscore how public health shocks can impact protective institutions tasked with safeguarding vulnerable populations.

This paper also highlights the critical role of institutional responsiveness during periods of heightened maltreatment risk. Enhancing flexibility in placement norms, so that placement decisions can adjust to changing risk conditions, would enable CPS to more effectively protect vulnerable children. Similarly, increasing the flexibility of state and local child welfare funding would allow agencies to scale resources in response to emerging crises. In particular, such funding could support timely hiring and training of staff, the expansion of

preventive services, and overall improvements in agency effectiveness. More broadly, aligning child protection efforts with real time changes in risk is essential to reducing the long term developmental and economic consequences of unaddressed maltreatment.

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# Appendix

## Child Protection in Response to Public Health Crises: Evidence from the Opioid Epidemic

Jeongsoo Suh

# A First Stage Regression

Table A1: First Stage Regressions

|                           | (1)<br>State         | (2)<br>State         | (3)<br>County        | (4)<br>County        |
|---------------------------|----------------------|----------------------|----------------------|----------------------|
| Exposure                  | -0.184***<br>(0.046) | -0.179***<br>(0.044) | -0.110***<br>(0.024) | -0.098***<br>(0.024) |
| Unemployment              |                      | 0.064*<br>(0.035)    |                      | 0.050***<br>(0.015)  |
| Labor Force Participation |                      | 0.044***<br>(0.012)  |                      | 0.014***<br>(0.004)  |
| Percent Black             |                      | 0.014**<br>(0.006)   |                      | 0.008***<br>(0.002)  |
| Percent Female            |                      | -0.117<br>(0.102)    |                      | -0.047**<br>(0.019)  |
| Percent Child             |                      | -0.070**<br>(0.027)  |                      | -0.017<br>(0.014)    |
| Constant                  | -0.156***<br>(0.044) | 4.266<br>(5.626)     | -0.118***<br>(0.024) | 1.189<br>(0.957)     |
| R-Squared                 | 0.330                | 0.550                | 0.066                | 0.177                |
| F Statistics              | 16.304               | 8.386                | 20.524               | 12.524               |
| Observations              | 49                   | 49                   | 643                  | 643                  |

**Notes.** This table presents point estimates and standard errors from weighted least squares (WLS) regressions of the change in OxyContin misuse rates from 2008 to 2012 on the exposure measures. Columns (1) and (2) are based on a state-level analysis. Columns (3) and (4) are based on a county-level analysis. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## B Covariates Orthogonality

Table A2: Covariate Balance

|                        | (1)<br>Unemployment<br>Rate | (2)<br>LFP<br>Rate  | (3)<br>Percent<br>Black | (4)<br>Percent<br>Female | (5)<br>Percent<br>Child |
|------------------------|-----------------------------|---------------------|-------------------------|--------------------------|-------------------------|
| <i>Panel A: State</i>  |                             |                     |                         |                          |                         |
| OxyContin Misuse       | 0.904<br>(0.942)            | 0.833<br>(0.817)    | 0.499<br>(0.566)        | -0.055<br>(0.079)        | -0.047<br>(0.279)       |
| Constant               | 2.698***<br>(0.670)         | -1.274**<br>(0.580) | -0.202<br>(0.402)       | -0.029<br>(0.056)        | -0.811***<br>(0.198)    |
| Observations           | 49                          | 49                  | 49                      | 49                       | 49                      |
| R-Squared              | 0.019                       | 0.022               | 0.016                   | 0.010                    | 0.001                   |
| <i>Panel B: County</i> |                             |                     |                         |                          |                         |
| Prescription Opioids   | 0.256<br>(0.375)            | 0.028<br>(0.378)    | 0.042<br>(0.063)        | -0.094***<br>(0.022)     | 0.132***<br>(0.039)     |
| Constant               | 4.239***<br>(0.499)         | -0.912**<br>(0.362) | 0.158*<br>(0.080)       | 0.074***<br>(0.025)      | -0.297***<br>(0.055)    |
| Observations           | 643                         | 643                 | 643                     | 643                      | 643                     |
| R-Squared              | 0.003                       | 0.000               | 0.001                   | 0.039                    | 0.068                   |

**Notes.** This table reports the results of regressing state and county covariates on pre-reformulation OxyContin misuse rate and Schedule II opioid prescriptions per capita. Standard errors in parentheses are clustered at the state level.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## C Supplemental Summary Statistics

Table A3: Summary Statistics for Maltreatment Allegations

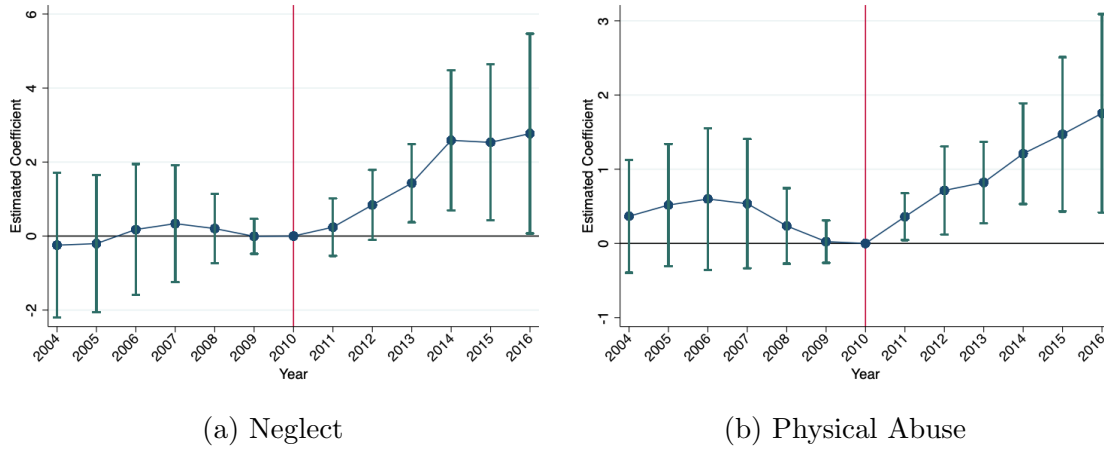
| Variable                              | All<br>Allegations | Professional<br>Allegations |
|---------------------------------------|--------------------|-----------------------------|
| <i>Panel A: Maltreatment Types</i>    |                    |                             |
| Neglect                               | 60.4               | 53.1                        |
| Physical Abuse                        | 21.8               | 24.1                        |
| Sexual Abuse                          | 7.5                | 8.4                         |
| Psychological/Emotional Maltreatment  | 6.9                | 7.0                         |
| Medical Neglect                       | 2.4                | 2.7                         |
| Other                                 | 9.5                | 9.1                         |
| <i>Panel B: Child Characteristics</i> |                    |                             |
| White                                 | 60.0               | 58.5                        |
| Black                                 | 25.4               | 26.0                        |
| Male                                  | 49.6               | 49.6                        |
| Female                                | 49.8               | 49.9                        |
| Young (age < 7 )                      | 45.8               | 43.3                        |
| Old (age ≥ 7 )                        | 53.5               | 56.3                        |
| <i>Panel C: Report Sources</i>        |                    |                             |
| Education Personnel                   | 16.7               | 29.6                        |
| Legal Personnel                       | 16.5               | 29.4                        |
| Social Services Personnel             | 10.2               | 18.2                        |
| Medical Personnel                     | 7.5                | 13.3                        |
| Mental Health Personnel               | 4.7                | 8.3                         |
| Child Daycare Provider                | 0.7                | 1.2                         |
| Relatives                             | 7.3                | -                           |
| Parent                                | 6.5                | -                           |
| Friends/Neighbors                     | 5.2                | -                           |
| Substitute Care Provider              | 0.4                | -                           |
| Alleged Victim                        | 0.4                | -                           |
| Alleged Perpetrator                   | 0.1                | -                           |
| Anonymous Reporter                    | 9.2                | -                           |
| Unknown                               | 14.6               | -                           |

**Notes.** This table reports the percentage of allegations by maltreatment types, child characteristics, and report sources, separately for all allegations and for allegations made by professionals during the sample period from 2004 to 2016. Professional allegations refer to those reported by education personnel, legal personnel, social services personnel, medical personnel, mental health providers, and child daycare providers.



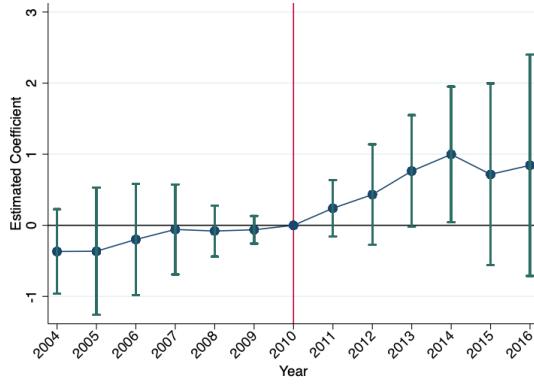
## D Allegations by Type of Maltreatment and Source of Report - State

Figure A1: Allegations by Type of Maltreatment - State

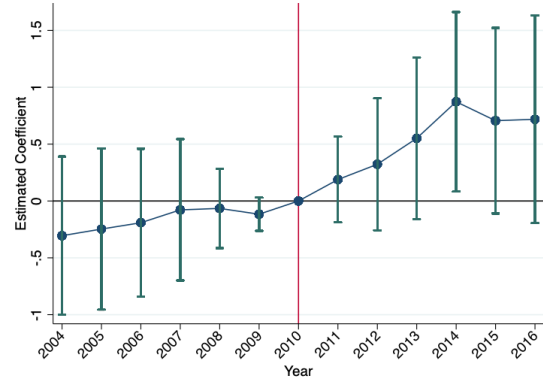


**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 that are adjusted for within-state clustering. Dependent variables are neglect and physical abuse per 1,000 children. Regressions are weighted by child population.

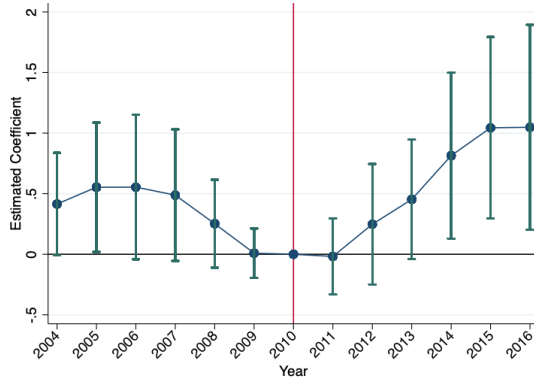
Figure A2: Allegations by Source of Report - State



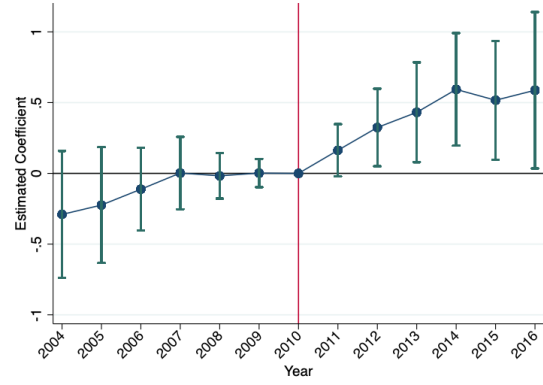
(a) Education Personnel



(b) Legal Personnel



(c) Social Services Personnel



(d) Medical Personnel

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 that are adjusted for within-state clustering. Dependent variables are maltreatment allegations per 1,000 children reported by education, legal, social services, and medical personnel. Regressions are weighted by child population.

Table A4: Allegations by Type of Maltreatment and Source of Report - State

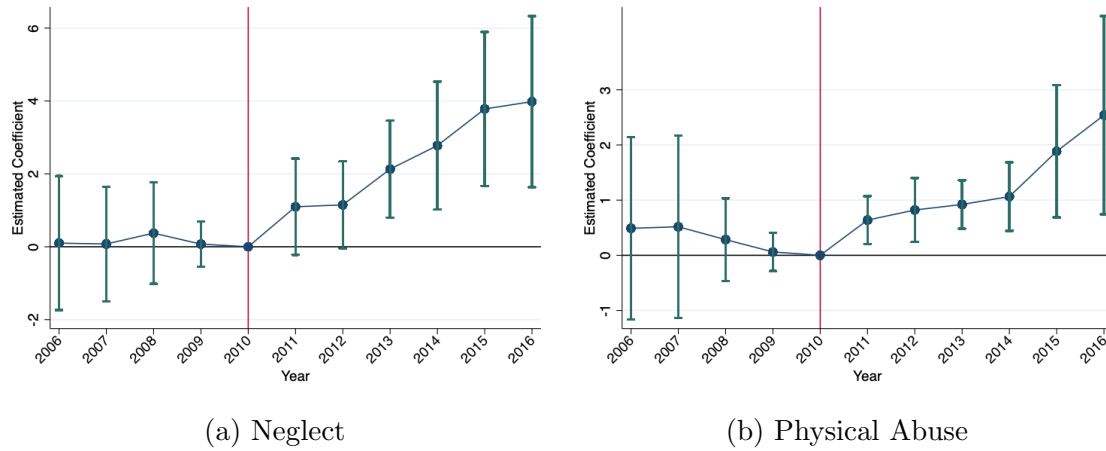
|                   | (1)                           | (2)                            | (3)                         | (4)                          | (5)                           | (6)                           |
|-------------------|-------------------------------|--------------------------------|-----------------------------|------------------------------|-------------------------------|-------------------------------|
|                   | Neglect                       | Physical                       | Education                   | Legal                        | Social Services               | Medical                       |
| Pre-reformulation | 0.051<br>(0.646)              | 0.366<br>(0.319)               | -0.178<br>(0.256)           | -0.160<br>(0.228)            | 0.365*<br>(0.194)             | -0.098<br>(0.117)             |
| Short-run         | 0.826**<br>(0.371)<br>[6.2%]  | 0.632***<br>(0.227)<br>[10.4%] | 0.472<br>(0.287)<br>[6.9%]  | 0.350<br>(0.264)<br>[4.9%]   | 0.228<br>(0.197)<br>[5.0%]    | 0.302**<br>(0.119)<br>[9.6%]  |
| Medium-run        | 2.604**<br>(1.090)<br>[19.7%] | 1.473***<br>(0.491)<br>[24.1%] | 0.840<br>(0.625)<br>[12.3%] | 0.758*<br>(0.411)<br>[10.6%] | 0.966**<br>(0.369)<br>[21.1%] | 0.557**<br>(0.218)<br>[17.7%] |
| Mean (2010)       | 13.237                        | 6.105                          | 6.839                       | 7.122                        | 4.58                          | 3.146                         |
| R-Squared         | 0.825                         | 0.683                          | 0.825                       | 0.919                        | 0.844                         | 0.850                         |
| Observations      | 634                           | 634                            | 634                         | 634                          | 634                           | 634                           |
| State FE          | Yes                           | Yes                            | Yes                         | Yes                          | Yes                           | Yes                           |
| Year FE           | Yes                           | Yes                            | Yes                         | Yes                          | Yes                           | Yes                           |
| $X_{st}$          | Yes                           | Yes                            | Yes                         | Yes                          | Yes                           | Yes                           |

**Notes.** This table reports point estimates and standard errors from Equation 3, where dependent variables are maltreatment allegations per 1,000 children, broken down by report type and source. Regressions are weighted by child population. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

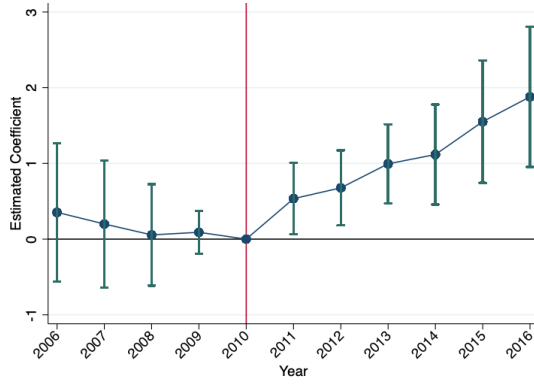
## E Allegations by Type of Maltreatment and Source of Report - County

Figure A3: Allegations by Type of Maltreatment - County

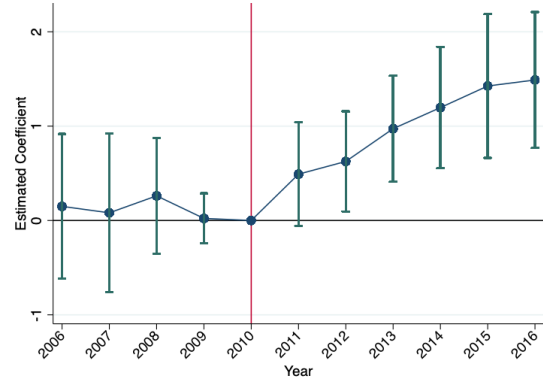


**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 2 that are adjusted for within-state clustering. Dependent variables are neglect and physical abuse per 1,000 children. Regressions are weighted by child population.

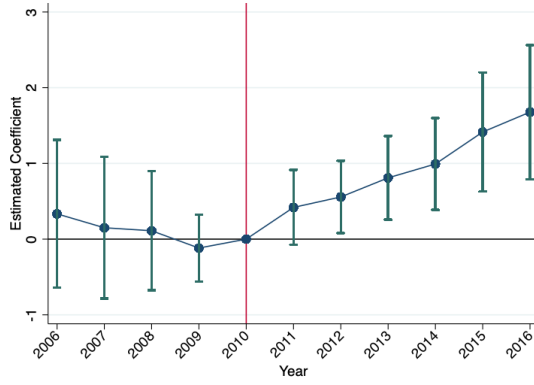
Figure A4: Allegations by Source of Report - County



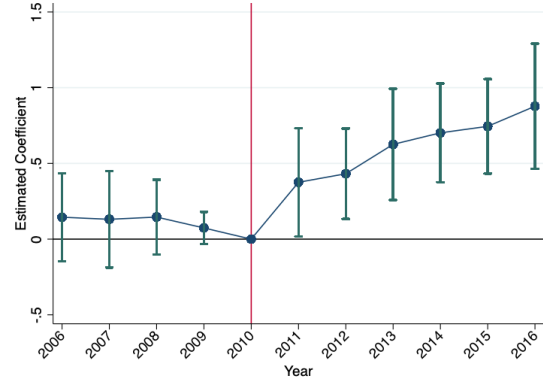
(a) Education Personnel



(b) Legal Personnel



(c) Social Services Personnel



(d) Medical Personnel

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 2 that are adjusted for within-state clustering. Dependent variables are maltreatment allegations per 1,000 children reported by education, legal, social services, and medical personnel. Regressions are weighted by child population.

Table A5: Allegations by Type of Maltreatment and Source of Report - County

|                   | (1)                            | (2)                            | (3)                            | (4)                            | (5)                            | (6)                            |
|-------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
|                   | Neglect                        | Physical                       | Education                      | Legal                          | Social Services                | Medical                        |
| Pre-reformulation | 0.161<br>(0.625)               | 0.333<br>(0.537)               | 0.173<br>(0.316)               | 0.130<br>(0.300)               | 0.116<br>(0.376)               | 0.124<br>(0.111)               |
| Short-run         | 1.456**<br>(0.604)<br>[11.0%]  | 0.792***<br>(0.221)<br>[13.0%] | 0.734***<br>(0.219)<br>[10.7%] | 0.694**<br>(0.262)<br>[9.7%]   | 0.594**<br>(0.243)<br>[13.0%]  | 0.477***<br>(0.166)<br>[15.2%] |
| Medium-run        | 3.491***<br>(0.948)<br>[26.4%] | 1.815***<br>(0.571)<br>[29.7%] | 1.507***<br>(0.372)<br>[22.0%] | 1.363***<br>(0.351)<br>[19.1%] | 1.354***<br>(0.363)<br>[29.6%] | 0.772***<br>(0.153)<br>[24.5%] |
| Mean (2010)       | 13.237                         | 6.105                          | 6.839                          | 7.122                          | 4.58                           | 3.146                          |
| R-Squared         | 0.852                          | 0.704                          | 0.807                          | 0.899                          | 0.836                          | 0.854                          |
| Observations      | 7073                           | 7073                           | 7073                           | 7073                           | 7073                           | 7073                           |
| County FE         | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |
| Year FE           | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |
| $X_{ct}$          | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |

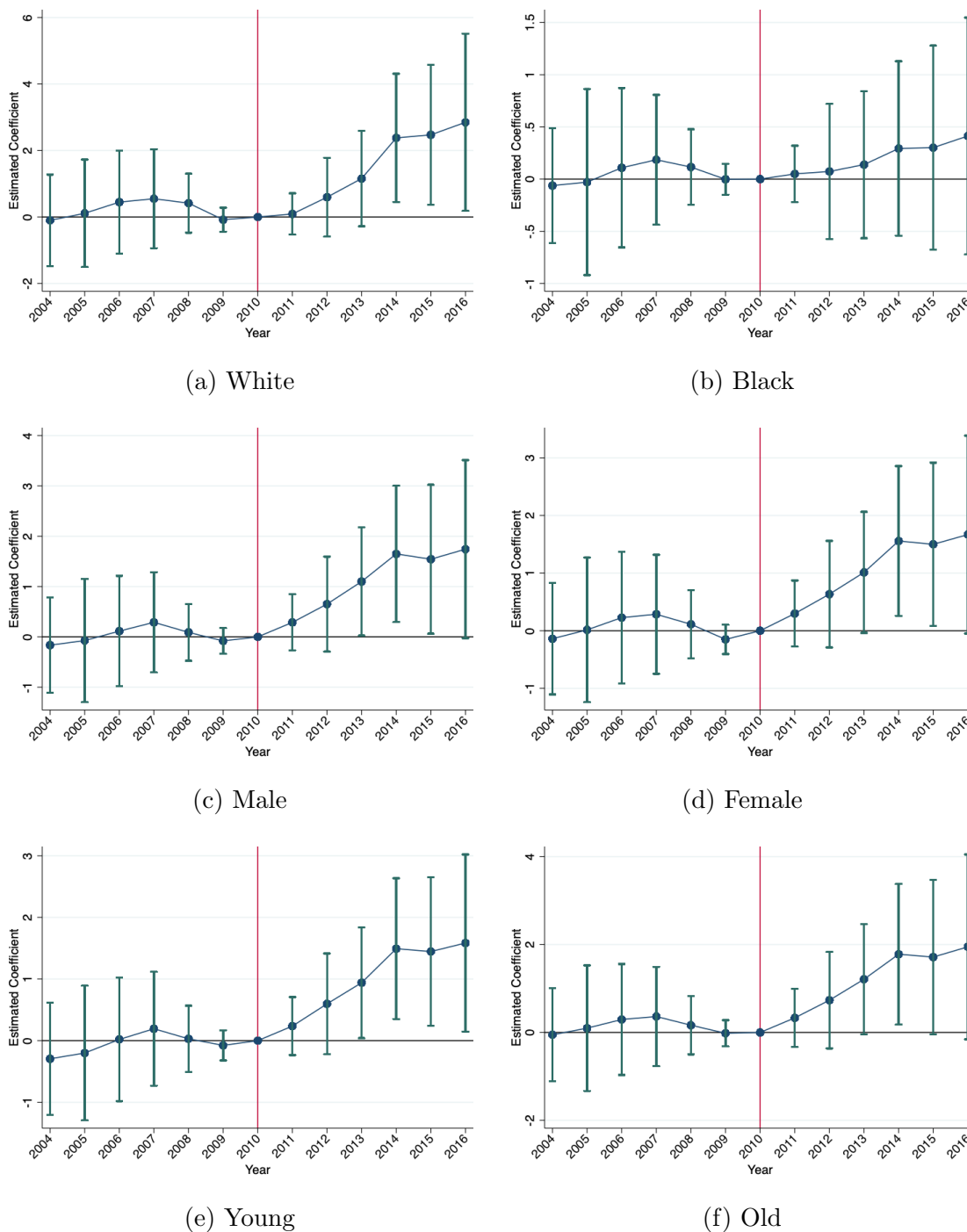
**Notes.** This table reports point estimates and standard errors from Equation 4, where dependent variables are maltreatment allegations per 1,000 children, broken down by report type and source. Regressions are weighted by child population. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$



## F Allegations by Child Characteristics - State

Figure A5: Maltreatment Allegations by Child Characteristics - State



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 that are adjusted for within-state clustering. Dependent variables are maltreatment allegations per 1,000 children, broken down by child characteristics. Regressions are weighted by child population.



Table A6: Maltreatment Allegations by Child Characteristics - State

|                   | (1)                           | (2)                        | (3)                           | (4)                           | (5)                           | (6)                           |
|-------------------|-------------------------------|----------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                   | White                         | Black                      | Male                          | Female                        | Young                         | Old                           |
| Pre-reformulation | 0.224<br>(0.531)              | 0.054<br>(0.246)           | 0.033<br>(0.364)              | 0.057<br>(0.378)              | -0.049<br>(0.348)             | 0.142<br>(0.423)              |
| Short-run         | 0.607<br>(0.479)<br>[4.4%]    | 0.086<br>(0.254)<br>[1.4%] | 0.673*<br>(0.390)<br>[5.7%]   | 0.644<br>(0.387)<br>[5.4%]    | 0.585*<br>(0.332)<br>[5.5%]   | 0.753<br>(0.455)<br>[5.7%]    |
| Medium-run        | 2.547**<br>(1.090)<br>[18.4%] | 0.332<br>(0.485)<br>[5.5%] | 1.629**<br>(0.751)<br>[13.7%] | 1.562**<br>(0.722)<br>[13.0%] | 1.491**<br>(0.612)<br>[14.1%] | 1.799**<br>(0.894)<br>[13.6%] |
| Mean (2010)       | 13.834                        | 6.047                      | 11.882                        | 11.975                        | 10.582                        | 13.271                        |
| R-Squared         | 0.864                         | 0.913                      | 0.870                         | 0.865                         | 0.877                         | 0.864                         |
| Observations      | 621                           | 621                        | 634                           | 634                           | 634                           | 634                           |
| State FE          | Yes                           | Yes                        | Yes                           | Yes                           | Yes                           | Yes                           |
| Year FE           | Yes                           | Yes                        | Yes                           | Yes                           | Yes                           | Yes                           |
| $X_{st}$          | Yes                           | Yes                        | Yes                           | Yes                           | Yes                           | Yes                           |

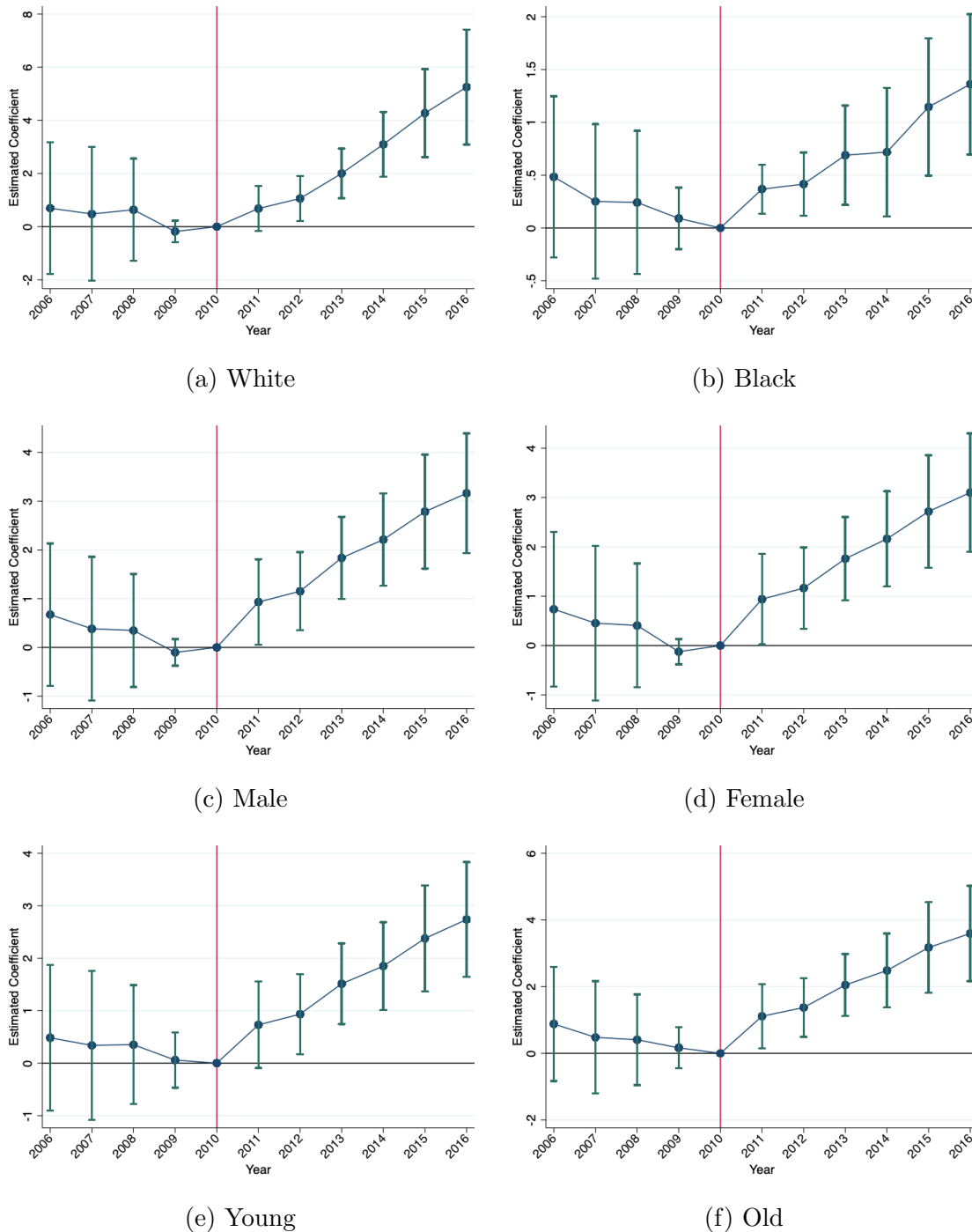
**Notes.** This table reports point estimates and standard errors from Equation 3, where the dependent variables are maltreatment allegations per 1,000 children, broken down by child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$



## G Allegations by Child Characteristics - County

Figure A6: Maltreatment Allegations by Child Characteristics - County



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 2 that are adjusted for within-state clustering. Dependent variables are maltreatment allegations per 1,000 children, broken down by child characteristics. Regressions are weighted by child population.

Table A7: Maltreatment Allegations by Child Characteristics - County

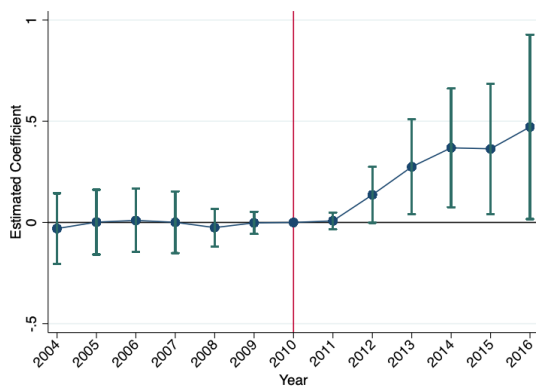
|                   | (1)                            | (2)                            | (3)                            | (4)                            | (5)                            | (6)                            |
|-------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
|                   | White                          | Black                          | Male                           | Female                         | Young                          | Old                            |
| Pre-reformulation | 0.403<br>(0.891)               | 0.264<br>(0.281)               | 0.319<br>(0.509)               | 0.362<br>(0.547)               | 0.306<br>(0.544)               | 0.479<br>(0.645)               |
| Short-run         | 1.244***<br>(0.339)<br>[9.0%]  | 0.490***<br>(0.137)<br>[8.1%]  | 1.304***<br>(0.391)<br>[11.0%] | 1.286***<br>(0.410)<br>[10.7%] | 1.057***<br>(0.374)<br>[10.0%] | 1.510***<br>(0.429)<br>[11.4%] |
| Medium-run        | 4.179***<br>(0.801)<br>[30.2%] | 1.068***<br>(0.300)<br>[17.7%] | 2.707***<br>(0.532)<br>[22.8%] | 2.649***<br>(0.524)<br>[22.1%] | 2.309***<br>(0.465)<br>[21.8%] | 3.070***<br>(0.620)<br>[23.1%] |
| Mean (2010)       | 13.834                         | 6.047                          | 11.882                         | 11.975                         | 10.582                         | 13.271                         |
| R-Squared         | 0.891                          | 0.918                          | 0.868                          | 0.864                          | 0.876                          | 0.856                          |
| Observations      | 6952                           | 6952                           | 7073                           | 7073                           | 7073                           | 7073                           |
| County FE         | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |
| Year FE           | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |
| $X_{ct}$          | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |

**Notes.** This table reports point estimates and standard errors from Equation 4, where the dependent variables are maltreatment allegations per 1,000 children, broken down by child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

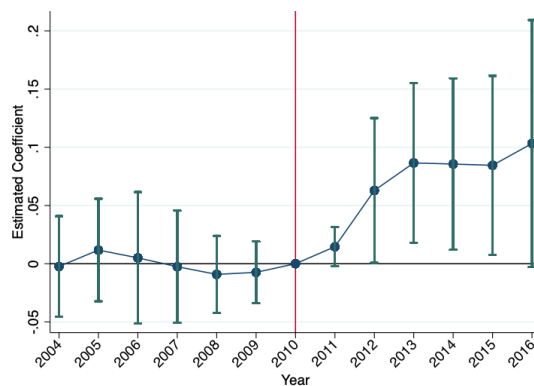
\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## H False Negatives by Type of Maltreatment and Child Characteristics - State

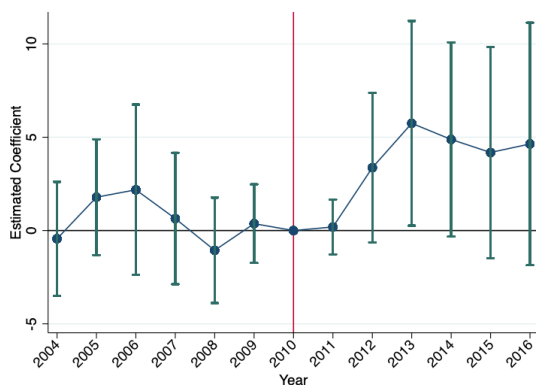
Figure A7: False Negative Rate by Type of Maltreatment - State



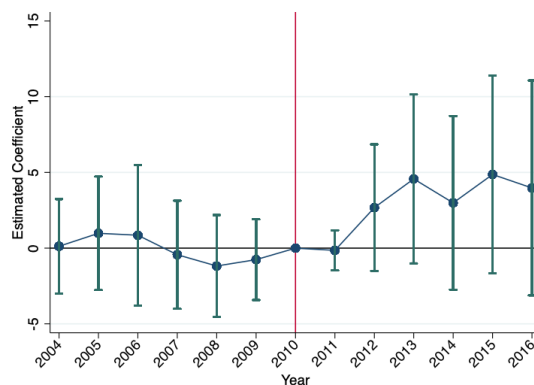
(a) Neglect, per 1,000 Children



(b) Physical Abuse, per 1,000 Children



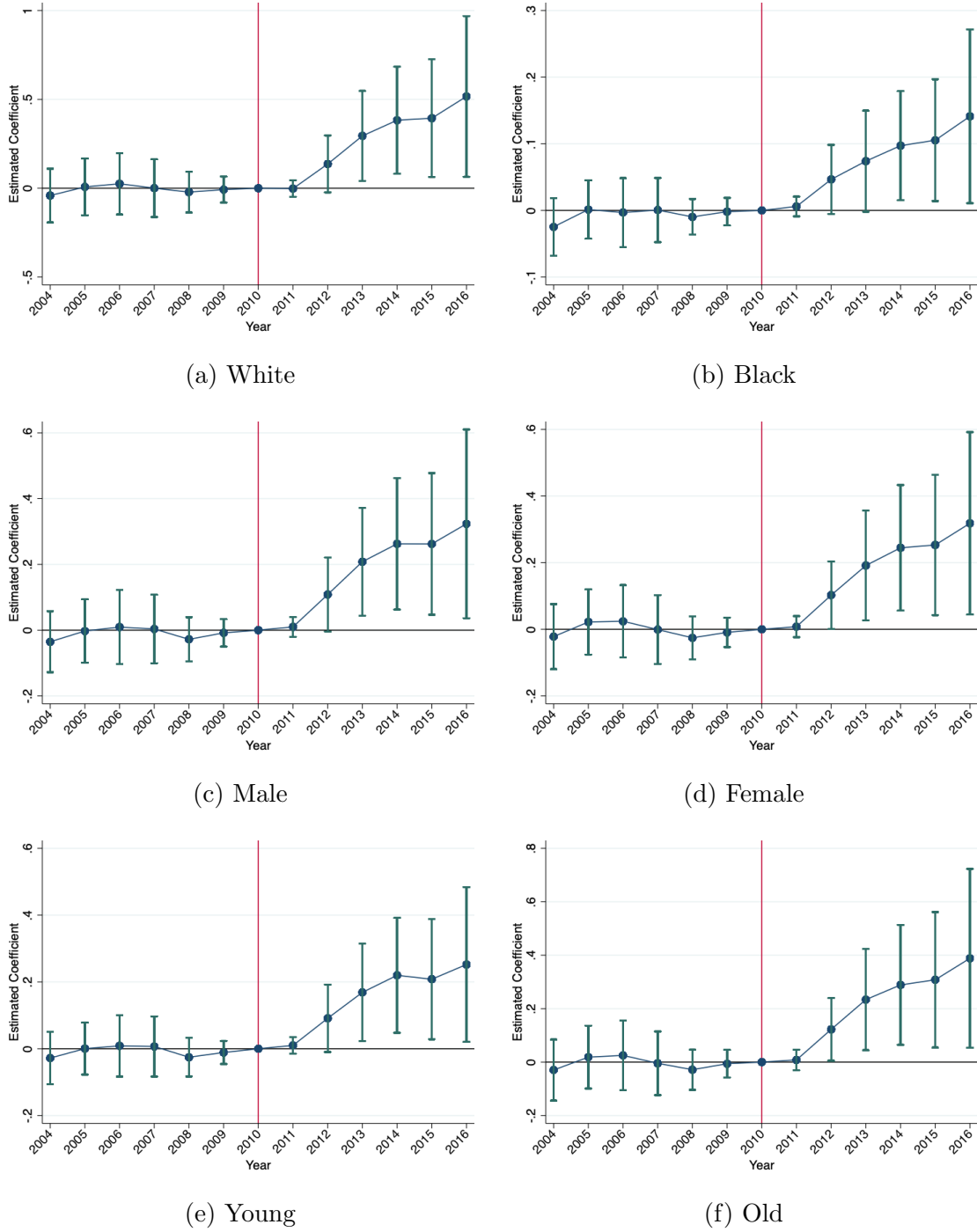
(c) Neglect, per 1,000 Allegations



(d) Physical Abuse, per 1,000 Allegations

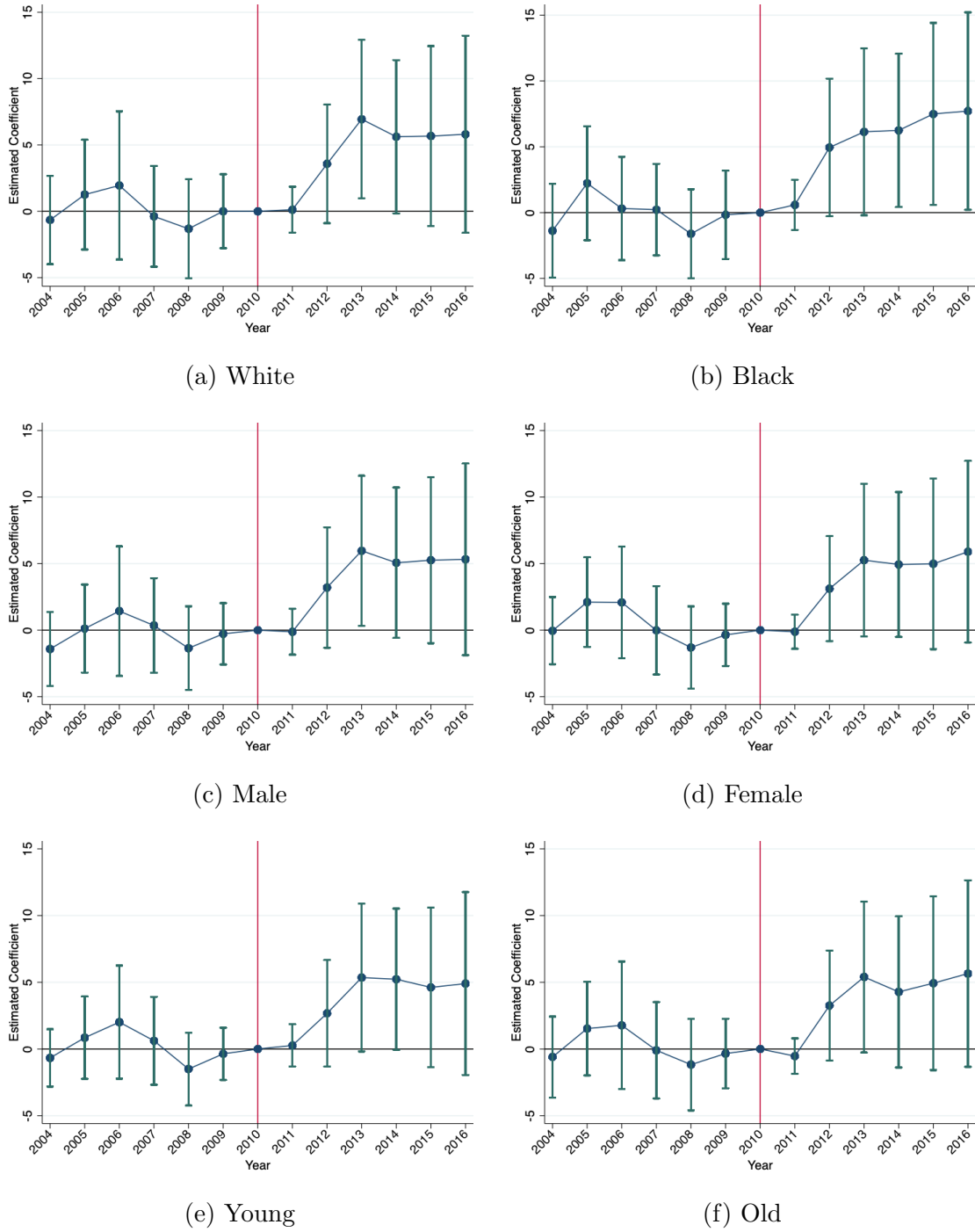
**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 that are adjusted for within-state clustering. Dependent variables are false negative rates by type of maltreatment. Regressions are weighted by child population.

Figure A8: False Negatives per 1,000 Children by Child Characteristics - State



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 1 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 children by child characteristics. Regressions are weighted by child population.

Figure A9: False Negatives per 1,000 Allegations by Child Characteristics - State



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 1 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 allegations by child characteristics. Regressions are weighted by child population.

Table A8: False Negatives per 1K Children by Type and Child Characteristics - State

|                   | (1)<br>Neglect                | (2)<br>Physical               | (3)<br>White                  | (4)<br>Black                  | (5)<br>Male                   | (6)<br>Female                 | (7)<br>Young                  | (8)<br>Old                    |
|-------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| Pre-reformulation | -0.009<br>(0.057)             | -0.001<br>(0.018)             | -0.008<br>(0.060)             | -0.007<br>(0.016)             | -0.011<br>(0.036)             | -0.004<br>(0.036)             | -0.009<br>(0.030)             | -0.006<br>(0.043)             |
| Short-run         | 0.137**<br>(0.060)<br>[17.7%] | 0.054**<br>(0.022)<br>[14.4%] | 0.141**<br>(0.069)<br>[16.0%] | 0.042*<br>(0.021)<br>[14.0%]  | 0.107**<br>(0.047)<br>[14.9%] | 0.099**<br>(0.044)<br>[13.8%] | 0.089**<br>(0.042)<br>[14.7%] | 0.120**<br>(0.050)<br>[14.4%] |
| Medium-run        | 0.394**<br>(0.172)<br>[51.0%] | 0.089**<br>(0.041)<br>[23.8%] | 0.422**<br>(0.174)<br>[47.8%] | 0.112**<br>(0.049)<br>[37.2%] | 0.277**<br>(0.112)<br>[38.6%] | 0.267**<br>(0.108)<br>[37.1%] | 0.222**<br>(0.093)<br>[36.8%] | 0.322**<br>(0.131)<br>[38.7%] |
| Mean (2010)       | 0.772                         | 0.374                         | 0.883                         | 0.301                         | 0.717                         | 0.719                         | 0.604                         | 0.831                         |
| R-Squared         | 0.705                         | 0.717                         | 0.750                         | 0.849                         | 0.738                         | 0.736                         | 0.741                         | 0.742                         |
| Observations      | 506                           | 506                           | 506                           | 506                           | 506                           | 506                           | 506                           | 506                           |
| State FE          | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           |
| Year FE           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           |
| $X_{st}$          | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           | Yes                           |

**Notes.** This table reports point estimates and standard errors from Equation 3, where the dependent variables are false negatives per 1,000 children by type of maltreatment and child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$



Table A9: False Negatives per 1K Allegations by Type and Child Characteristics - State

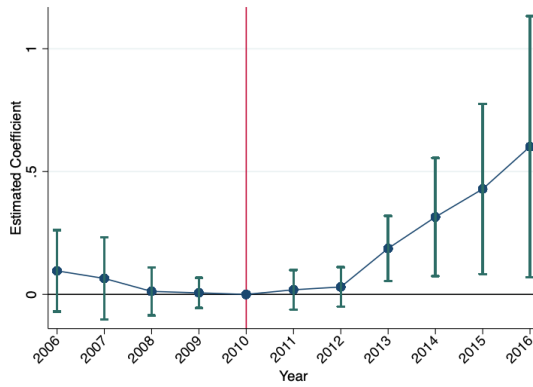
|                   | (1)                         | (2)                        | (3)                         | (4)                           | (5)                        | (6)                         | (7)                        | (8)                        |
|-------------------|-----------------------------|----------------------------|-----------------------------|-------------------------------|----------------------------|-----------------------------|----------------------------|----------------------------|
|                   | Neglect                     | Physical                   | White                       | Black                         | Male                       | Female                      | Young                      | Old                        |
| Pre-reformulation | 0.526<br>(1.262)            | -0.145<br>(1.528)          | 0.087<br>(1.665)            | -0.129<br>(1.370)             | -0.234<br>(1.421)          | 0.324<br>(1.267)            | 0.093<br>(1.145)           | 0.102<br>(1.503)           |
| Short-run         | 3.070*<br>(1.621)<br>[5.1%] | 2.346<br>(1.700)<br>[3.8%] | 3.504*<br>(1.792)<br>[5.4%] | 3.852*<br>(1.970)<br>[6.1%]   | 2.973<br>(1.779)<br>[4.8%] | 2.731<br>(1.647)<br>[4.4%]  | 2.740<br>(1.705)<br>[4.7%] | 2.675<br>(1.608)<br>[4.1%] |
| Medium-run        | 4.424<br>(2.738)<br>[7.3%]  | 3.838<br>(3.079)<br>[6.3%] | 5.529*<br>(3.152)<br>[8.5%] | 6.992**<br>(3.214)<br>[11.0%] | 5.039<br>(3.041)<br>[8.1%] | 5.149*<br>(2.956)<br>[8.3%] | 4.788<br>(2.882)<br>[8.2%] | 4.801<br>(3.031)<br>[7.4%] |
| Mean (2010)       | 60.588                      | 61.073                     | 65.2                        | 63.412                        | 62.356                     | 61.72                       | 58.223                     | 65.254                     |
| R-Squared         | 0.750                       | 0.730                      | 0.760                       | 0.717                         | 0.761                      | 0.768                       | 0.765                      | 0.756                      |
| Observations      | 506                         | 506                        | 506                         | 506                           | 506                        | 506                         | 506                        | 506                        |
| State FE          | Yes                         | Yes                        | Yes                         | Yes                           | Yes                        | Yes                         | Yes                        | Yes                        |
| Year FE           | Yes                         | Yes                        | Yes                         | Yes                           | Yes                        | Yes                         | Yes                        | Yes                        |
| $X_{st}$          | Yes                         | Yes                        | Yes                         | Yes                           | Yes                        | Yes                         | Yes                        | Yes                        |

**Notes.** This table reports point estimates and standard errors from Equation 3, where the dependent variables are false negatives per 1,000 children by maltreatment types and child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

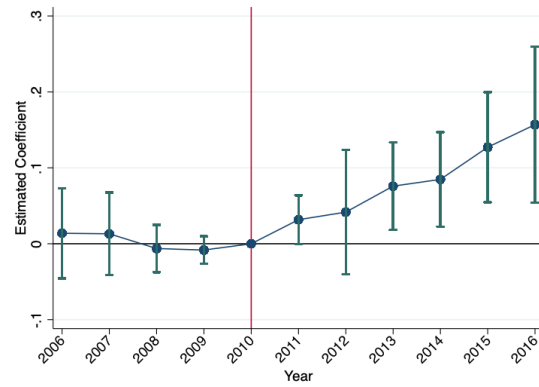
\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

# I False Negatives by Type of Maltreatment and Child Characteristics - County

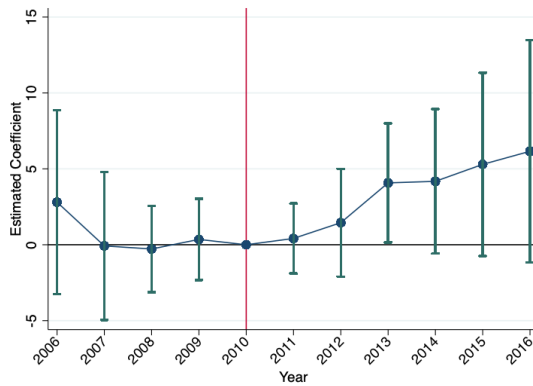
Figure A10: False Negative Rate by Maltreatment Type - County



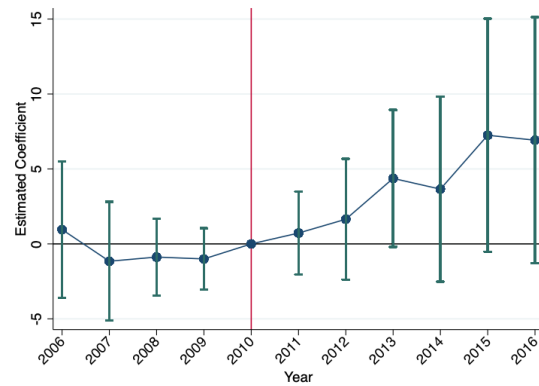
(a) Neglect, per 1,000 Children



(b) Physical Abuse, per 1,000 Children



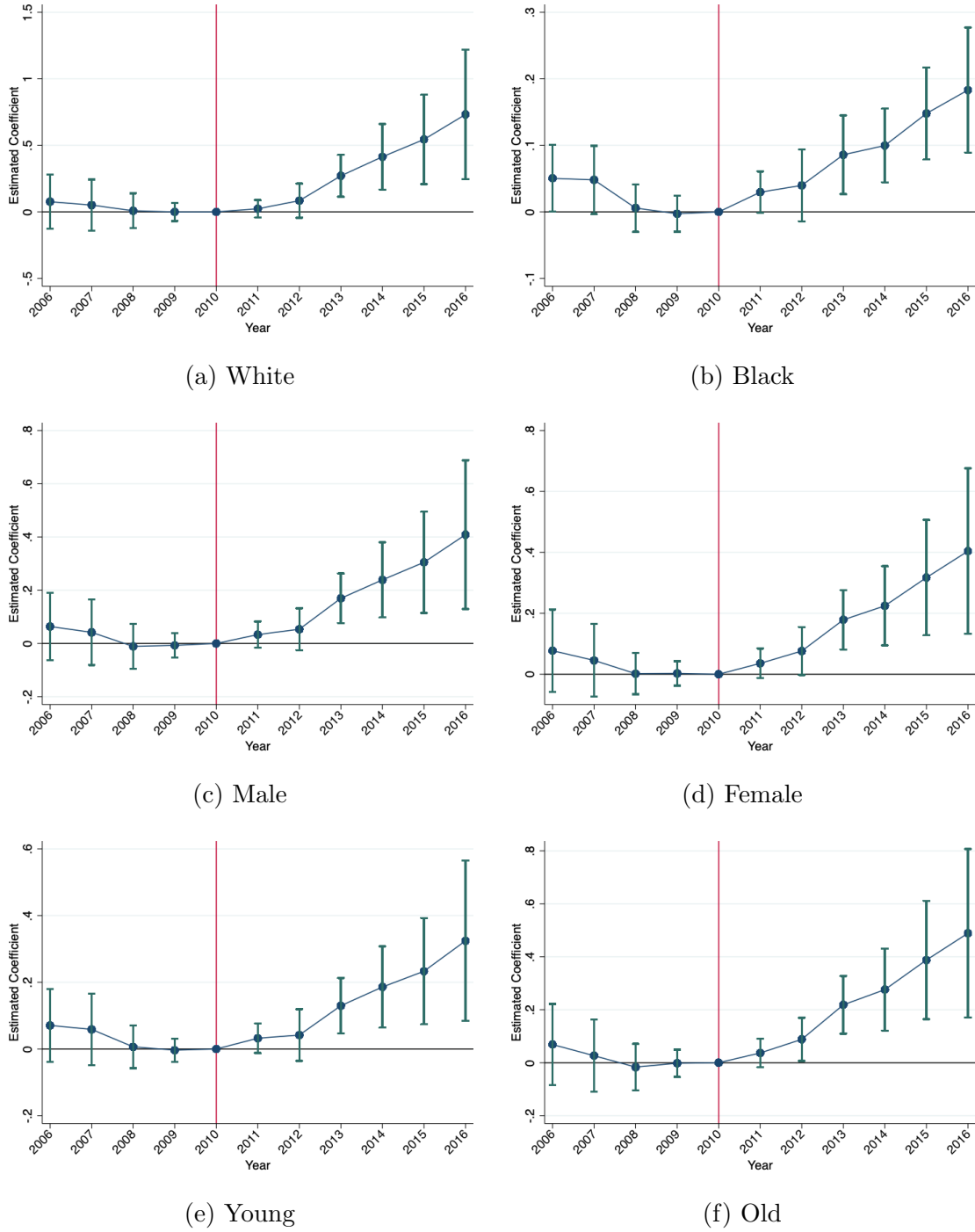
(c) Neglect, per 1,000 Allegations



(d) Physical Abuse, per 1,000 Allegations

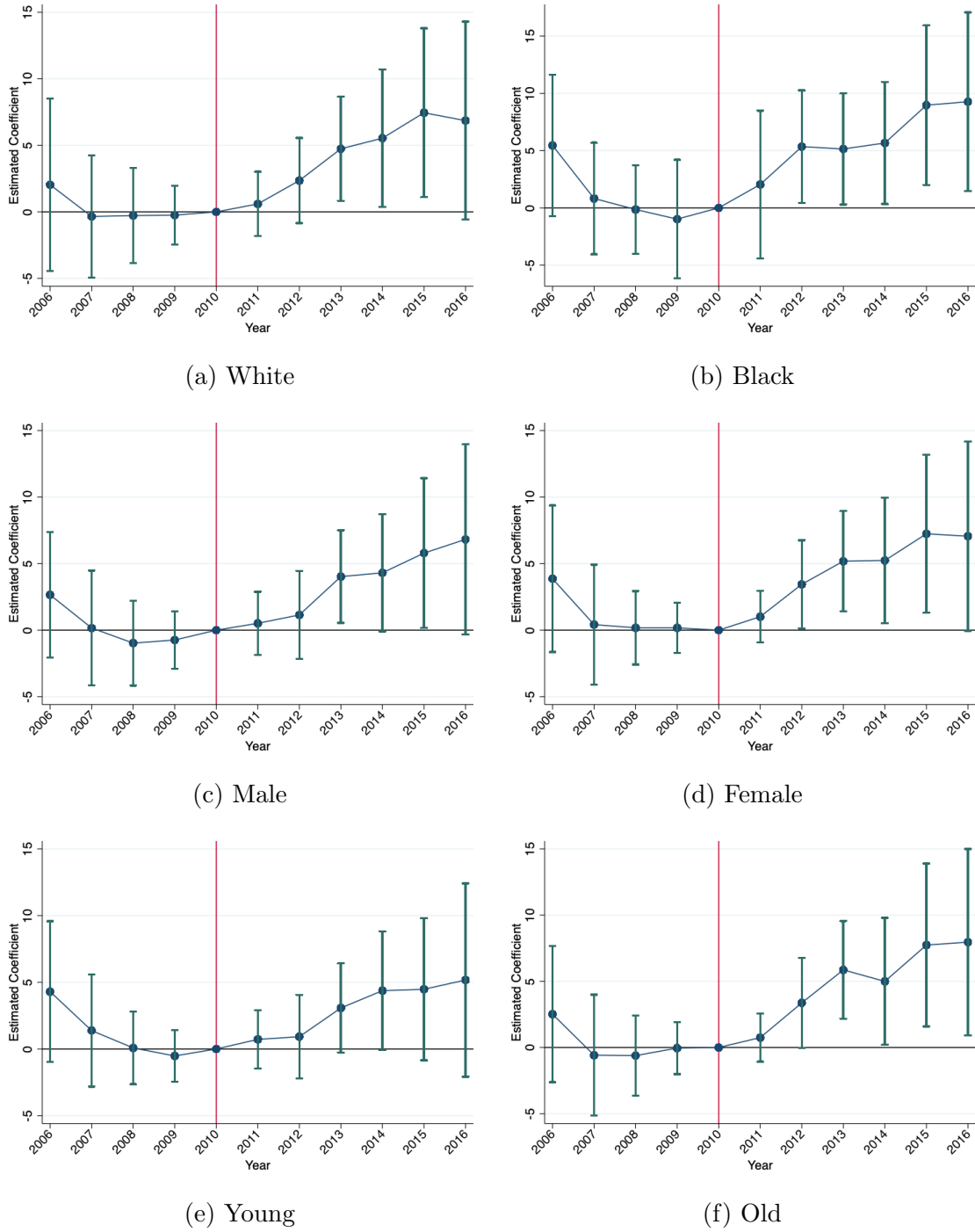
**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 2 that are adjusted for within-state clustering. Dependent variables are false negative rates by type of maltreatment. Regressions are weighted by child population.

Figure A11: False Negatives per 1,000 Children by Child Characteristics - County



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 2 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 children by child characteristics. Regressions are weighted by child population.

Figure A12: False Negatives per 1,000 Allegations by Child Characteristics - County



**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 2 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 allegations by child characteristics. Regressions are weighted by child population.

Table A10: False Negatives per 1K Children by Type and Child Characteristics - County

|                   | (1)                           | (2)                            | (3)                            | (4)                            | (5)                            | (6)                            | (7)                            | (8)                            |
|-------------------|-------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
|                   | Neglect                       | Physical                       | White                          | Black                          | Male                           | Female                         | Young                          | Old                            |
| Pre-reformulation | 0.043<br>(0.054)              | 0.003<br>(0.018)               | 0.032<br>(0.068)               | 0.024<br>(0.016)               | 0.020<br>(0.042)               | 0.030<br>(0.040)               | 0.032<br>(0.034)               | 0.018<br>(0.048)               |
| Short-run         | 0.078**<br>(0.033)<br>[10.1%] | 0.049*<br>(0.025)<br>[13.1%]   | 0.124***<br>(0.045)<br>[14.0%] | 0.052**<br>(0.020)<br>[17.3%]  | 0.085***<br>(0.029)<br>[11.9%] | 0.096***<br>(0.028)<br>[13.4%] | 0.068**<br>(0.028)<br>[11.3%]  | 0.113***<br>(0.029)<br>[13.6%] |
| Medium-run        | 0.444**<br>(0.186)<br>[57.5%] | 0.122***<br>(0.036)<br>[32.6%] | 0.556***<br>(0.177)<br>[63.0%] | 0.142***<br>(0.034)<br>[47.2%] | 0.314***<br>(0.100)<br>[43.8%] | 0.311***<br>(0.097)<br>[43.3%] | 0.245***<br>(0.085)<br>[40.6%] | 0.379***<br>(0.115)<br>[45.6%] |
| Mean (2010)       | 0.772                         | 0.374                          | 0.883                          | 0.301                          | 0.717                          | 0.719                          | 0.604                          | 0.831                          |
| R-Squared         | 0.684                         | 0.675                          | 0.732                          | 0.802                          | 0.701                          | 0.703                          | 0.701                          | 0.70                           |
| Observations      | 4587                          | 4587                           | 4587                           | 4587                           | 4587                           | 4587                           | 4587                           | 4587                           |
| County FE         | Yes                           | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |
| Year FE           | Yes                           | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |
| $X_{ct}$          | Yes                           | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            | Yes                            |

**Notes.** This table reports point estimates and standard errors from Equation 3, where the dependent variables are false negatives per 1,000 children by type of maltreatment and child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

Table A11: False Negatives per 1K Allegations by Type and Child Characteristics - County

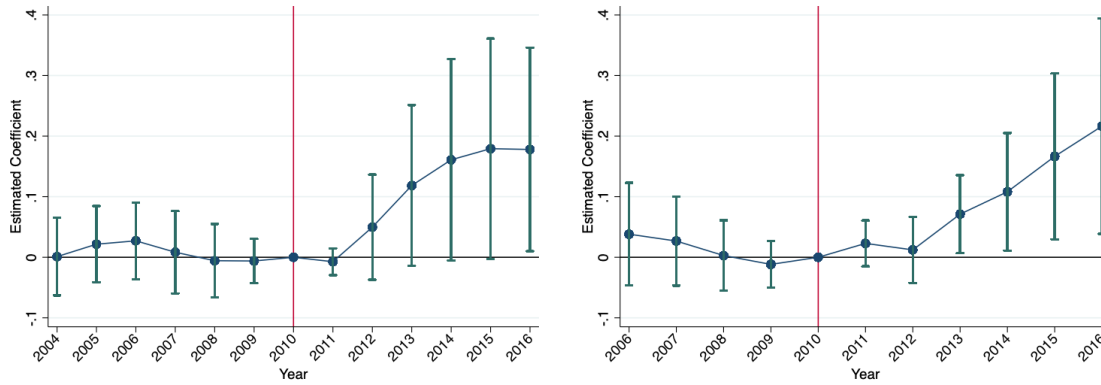
|                   | (1)                         | (2)                        | (3)                           | (4)                           | (5)                         | (6)                           | (7)                         | (8)                           |
|-------------------|-----------------------------|----------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------------|-----------------------------|-------------------------------|
|                   | Neglect                     | Physical                   | White                         | Black                         | Male                        | Female                        | Young                       | Old                           |
| Pre-reformulation | 0.662<br>(1.770)            | -0.551<br>(1.111)          | 0.262<br>(1.830)              | 1.204<br>(1.677)              | 0.215<br>(1.474)            | 1.103<br>(1.536)              | 1.228<br>(1.386)            | 0.280<br>(1.551)              |
| Short-run         | 1.953<br>(1.357)<br>[3.2%]  | 2.220<br>(1.668)<br>[3.6%] | 2.527*<br>(1.254)<br>[3.9%]   | 4.140**<br>(1.729)<br>[6.5%]  | 1.881<br>(1.263)<br>[3.0%]  | 3.180**<br>(1.210)<br>[5.2%]  | 1.577<br>(1.180)<br>[2.7%]  | 3.275***<br>(1.159)<br>[5.0%] |
| Medium-run        | 5.124*<br>(2.922)<br>[8.5%] | 5.848<br>(3.631)<br>[9.6%] | 6.520**<br>(2.964)<br>[10.0%] | 7.876**<br>(3.137)<br>[12.4%] | 5.568*<br>(2.794)<br>[8.9%] | 6.418**<br>(2.868)<br>[10.4%] | 4.645*<br>(2.743)<br>[8.0%] | 6.764**<br>(2.935)<br>[10.4%] |
| Mean (2010)       | 60.588                      | 61.073                     | 65.2                          | 63.412                        | 62.356                      | 61.72                         | 58.223                      | 65.254                        |
| R-Squared         | 0.641                       | 0.568                      | 0.669                         | 0.361                         | 0.672                       | 0.672                         | 0.678                       | 0.648                         |
| Observations      | 4587                        | 4587                       | 4587                          | 4587                          | 4587                        | 4587                          | 4587                        | 4587                          |
| County FE         | Yes                         | Yes                        | Yes                           | Yes                           | Yes                         | Yes                           | Yes                         | Yes                           |
| Year FE           | Yes                         | Yes                        | Yes                           | Yes                           | Yes                         | Yes                           | Yes                         | Yes                           |
| $X_{ct}$          | Yes                         | Yes                        | Yes                           | Yes                           | Yes                         | Yes                           | Yes                         | Yes                           |

**Notes.** This table reports point estimates and standard errors from Equation 4, where the dependent variables are maltreatment allegations per 1,000 allegations, broken down by child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

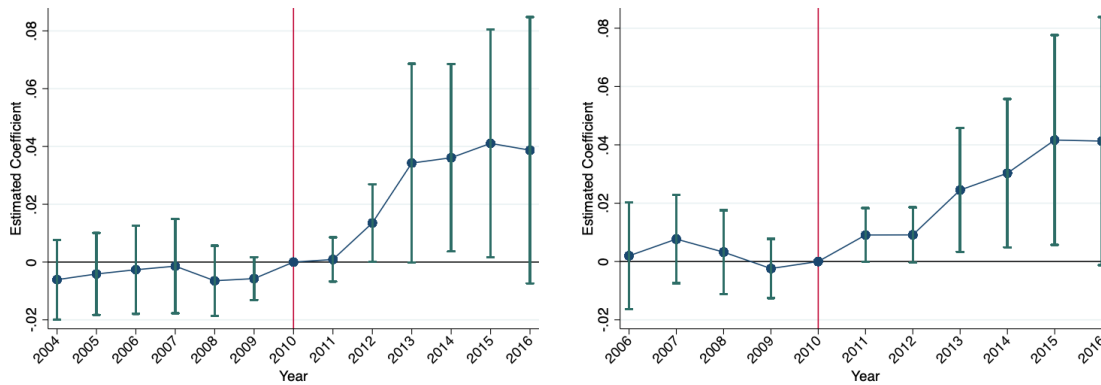
## J Alternative Measures of False Negatives

Figure A13: False Negatives per 1,000 Children - Alternative Measures



(a) State, Substantiation

(b) County, Substantiation

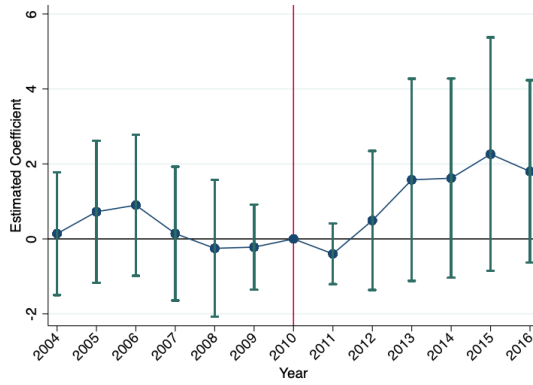


(c) State, Foster Care

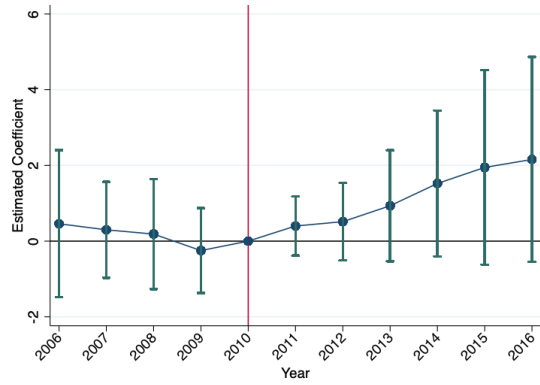
(d) County, Foster Care

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 children based on alternative measures. Regressions are weighted by child population.

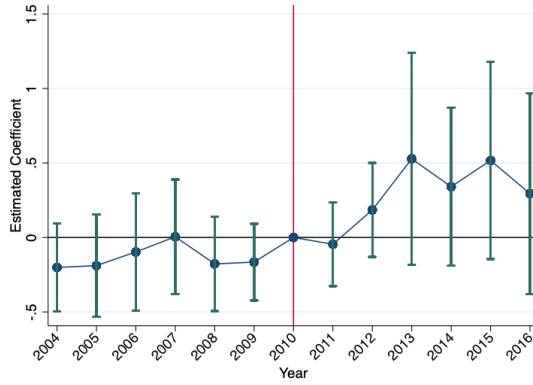
Figure A14: False Negatives per 1,000 Allegations - Alternative Measures



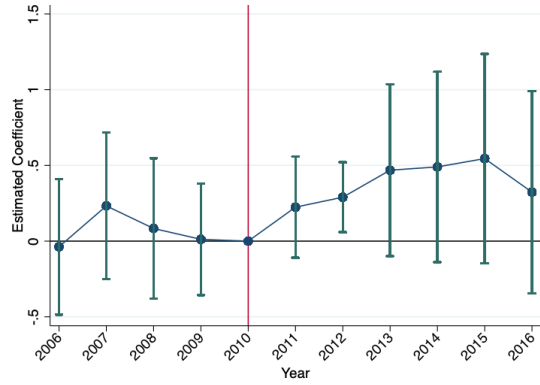
(a) State, Substantiation



(b) County, Substantiation



(c) State, Foster Care



(d) County, Foster Care

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 allegations based on alternative measures. Regressions are weighted by child population.



Table A12: Difference-in-Differences Results for False Negatives per 1K Children by Alternative Measures

|                      | (1)<br>State<br>Substantiation | (2)<br>State<br>Foster Care  | (3)<br>County<br>Substantiation | (4)<br>County<br>Foster Care   |
|----------------------|--------------------------------|------------------------------|---------------------------------|--------------------------------|
| Pre-reformulation    | 0.007<br>(0.026)               | -0.005<br>(0.006)            | 0.013<br>(0.029)                | 0.003<br>(0.006)               |
| Short-run            | 0.053<br>(0.038)<br>[14.0%]    | 0.016*<br>(0.008)<br>[18.0%] | 0.035*<br>(0.020)<br>[9.3%]     | 0.014***<br>(0.005)<br>[15.7%] |
| Medium-run           | 0.170**<br>(0.082)<br>[45.0%]  | 0.038*<br>(0.019)<br>[42.7%] | 0.162**<br>(0.067)<br>[42.9%]   | 0.037**<br>(0.016)<br>[41.6%]  |
| Mean (2010)          | 0.378                          | 0.089                        | 0.378                           | 0.089                          |
| R-Squared            | 0.752                          | 0.679                        | 0.699                           | 0.560                          |
| Observations         | 506                            | 506                          | 4587                            | 4587                           |
| State of County FE   | Yes                            | Yes                          | Yes                             | Yes                            |
| Year FE              | Yes                            | Yes                          | Yes                             | Yes                            |
| $X_{st}$ or $X_{ct}$ | Yes                            | Yes                          | Yes                             | Yes                            |

**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where the dependent variables are false negatives per 1,000 children by alternative measures. Columns (1) and (3) report the results for false negatives based on subsequent substantiation following the reports from professionals. Columns (2) and (4) report the results for false negatives based on subsequent foster care placements following the reports from professionals. Columns (1) and (2) are state-level results whereas Columns (3) and (4) are county-level results. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

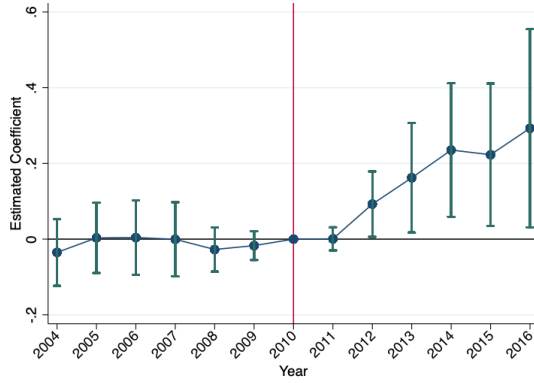
Table A13: Difference-in-Differences Results for False Negatives per 1K Allegations by Alternative Measures

|                      | (1)<br>State<br>Substantiation | (2)<br>State<br>Foster Care | (3)<br>County<br>Substantiation | (4)<br>County<br>Foster Care |
|----------------------|--------------------------------|-----------------------------|---------------------------------|------------------------------|
| Pre-reformulation    | 0.201<br>(0.738)               | -0.140<br>(0.133)           | 0.162<br>(0.641)                | 0.073<br>(0.181)             |
| Short-run            | 0.551<br>(0.848)<br>[2.0%]     | 0.218<br>(0.189)<br>[3.3%]  | 0.614<br>(0.420)<br>[2.2%]      | 0.325**<br>(0.152)<br>[4.9%] |
| Medium-run           | 1.852<br>(1.306)<br>[6.8%]     | 0.367<br>(0.291)<br>[5.5%]  | 1.863<br>(1.176)<br>[6.8%]      | 0.449<br>(0.303)<br>[6.7%]   |
| Mean (2010)          | 27.43                          | 6.68                        | 27.43                           | 6.68                         |
| R-Squared            | 0.864                          | 0.740                       | 0.779                           | 0.521                        |
| Observations         | 506                            | 506                         | 4587                            | 4587                         |
| State or County FE   | Yes                            | Yes                         | Yes                             | Yes                          |
| Year FE              | Yes                            | Yes                         | Yes                             | Yes                          |
| $X_{st}$ or $X_{ct}$ | Yes                            | Yes                         | Yes                             | Yes                          |

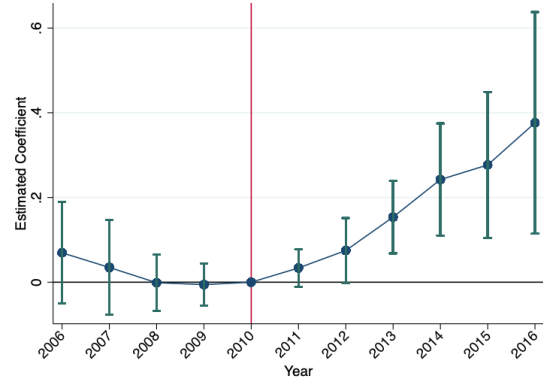
**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where The dependent variables are false negatives per 1,000 allegations by alternative measures. Columns (1) and (3) report the results for false negatives based on subsequent substantiation following the reports from professionals. Columns (2) and (4) report the results for false negatives based on subsequent foster care placements following the reports from professionals. Columns (1) and (2) are state-level results whereas Columns (3) and (4) are county-level results. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

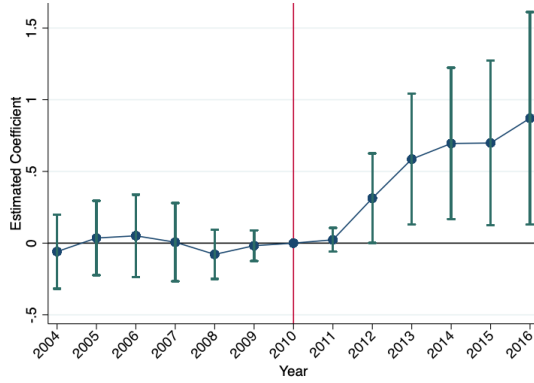
Figure A15: False Negatives per 1,000 Children - Alternative Time Frames



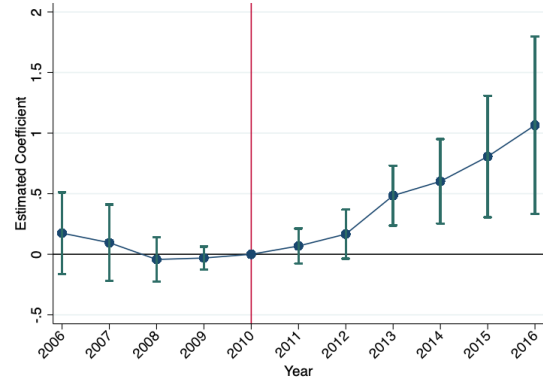
(a) State, 3 Months



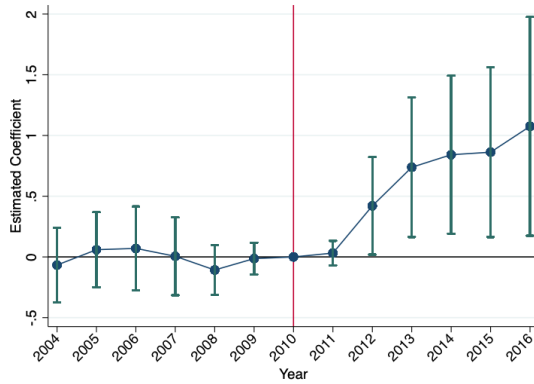
(b) County, 3 Months



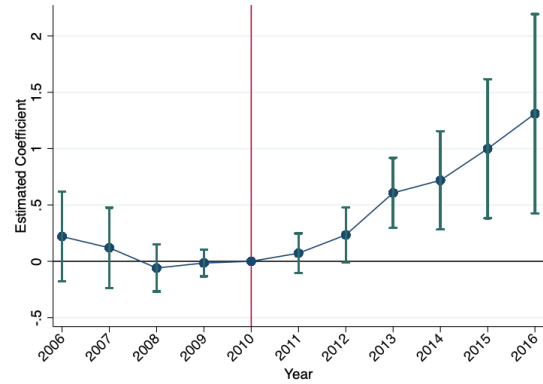
(c) State, 9 Months



(d) County, 9 Months



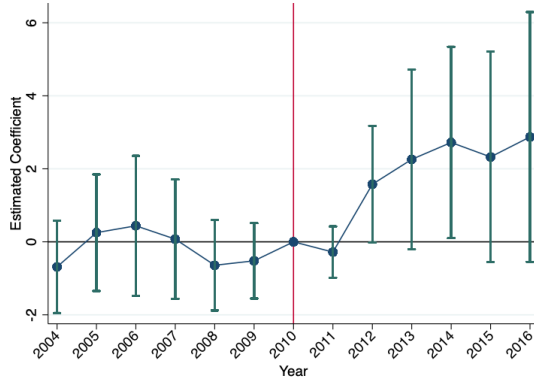
(e) State, 12 Months



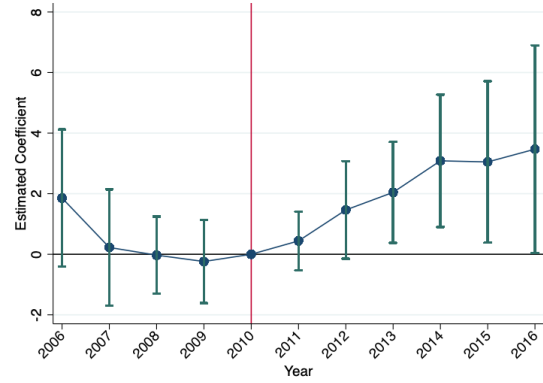
(f) County, 12 Months

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 children based on alternative measures. Regressions are weighted by child population.

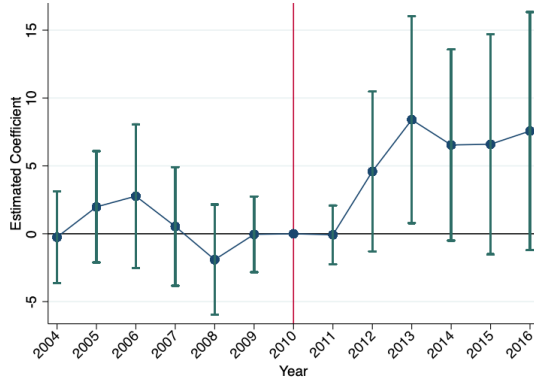
Figure A16: False Negatives per 1,000 Allegations - Alternative Time Frames



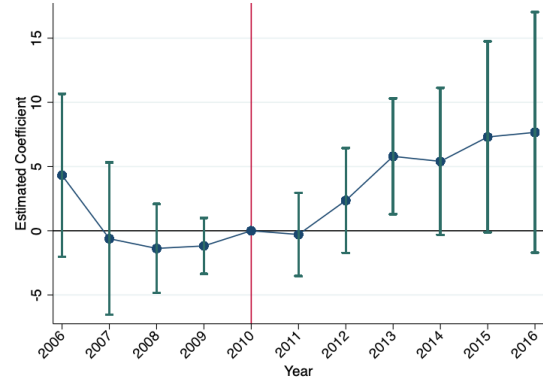
(a) State, 3 Months



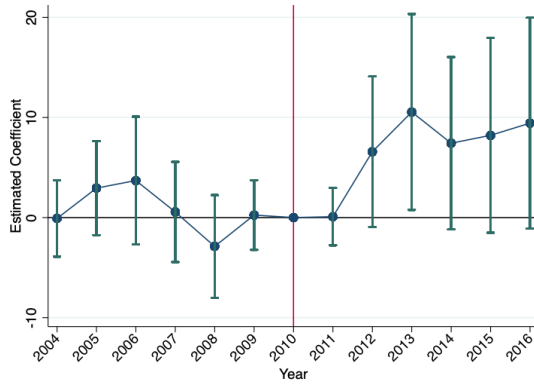
(b) County, 3 Months



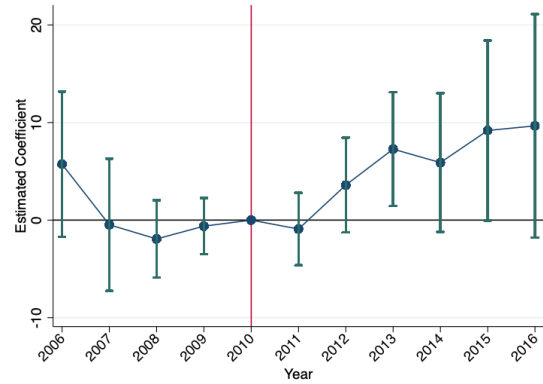
(c) State, 9 Months



(d) County, 9 Months



(e) State, 12 Months



(f) County, 12 Months

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2009) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Dependent variables are false negatives per 1,000 allegations based on alternative measures. Regressions are weighted by child population.

Table A14: Difference-in-Differences Results for False Negatives per 1K Children by Time Frame

|                      | (1)<br>State<br>3 Months      | (2)<br>State<br>9 Months      | (3)<br>State<br>12 Months     | (4)<br>County<br>3 Months      | (5)<br>County<br>9 Months      | (6)<br>County<br>12 Months     |
|----------------------|-------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|--------------------------------|
| Pre-reformulation    | -0.013<br>(0.034)             | -0.014<br>(0.094)             | -0.014<br>(0.111)             | 0.023<br>(0.037)               | 0.045<br>(0.103)               | 0.061<br>(0.117)               |
| Short-run            | 0.084**<br>(0.039)<br>[14.7%] | 0.303**<br>(0.127)<br>[14.8%] | 0.391**<br>(0.162)<br>[15.1%] | 0.086***<br>(0.026)<br>[15.1%] | 0.237***<br>(0.073)<br>[11.5%] | 0.300***<br>(0.091)<br>[11.6%] |
| Medium-run           | 0.246**<br>(0.101)<br>[43.2%] | 0.739**<br>(0.294)<br>[36.0%] | 0.906**<br>(0.359)<br>[34.9%] | 0.295***<br>(0.093)<br>[51.8%] | 0.813***<br>(0.260)<br>[39.6%] | 0.995***<br>(0.319)<br>[38.4%] |
| Mean (2010)          | 0.57                          | 2.052                         | 2.593                         | 0.57                           | 2.052                          | 2.593                          |
| R-Squared            | 0.718                         | 0.744                         | 0.750                         | 0.678                          | 0.719                          | 0.72                           |
| Observations         | 506                           | 506                           | 506                           | 4587                           | 4587                           | 4587                           |
| State of County FE   | Yes                           | Yes                           | Yes                           | Yes                            | Yes                            | Yes                            |
| Year FE              | Yes                           | Yes                           | Yes                           | Yes                            | Yes                            | Yes                            |
| $X_{st}$ or $X_{ct}$ | Yes                           | Yes                           | Yes                           | Yes                            | Yes                            | Yes                            |

**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where the dependent variables are false negatives per 1,000 children by the duration between the placement decision and re-referral. Columns (1), (2), and (3) are state-level results whereas Columns (4), (5), and (6) are county-level results. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

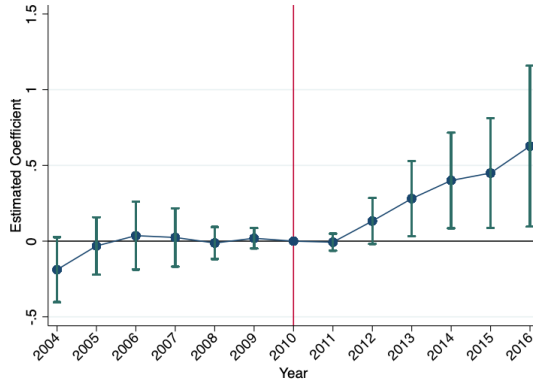
Table A15: Difference-in-Differences Results for False Negatives per 1K Allegations by Time Frame

|                      | (1)<br>State<br>3 Months     | (2)<br>State<br>9 Months    | (3)<br>State<br>12 Months   | (4)<br>County<br>3 Months     | (5)<br>County<br>9 Months   | (6)<br>County<br>12 Months  |
|----------------------|------------------------------|-----------------------------|-----------------------------|-------------------------------|-----------------------------|-----------------------------|
| Pre-reformulation    | -0.209<br>(0.608)            | 0.413<br>(1.648)            | 0.603<br>(1.923)            | 0.422<br>(0.673)              | 0.199<br>(1.786)            | 0.570<br>(1.995)            |
| Short-run            | 1.170<br>(0.714)<br>[4.8%]   | 4.258*<br>(2.327)<br>[4.8%] | 5.684*<br>(3.017)<br>[5.1%] | 1.304**<br>(0.581)<br>[5.3%]  | 2.575*<br>(1.466)<br>[2.9%] | 3.249*<br>(1.774)<br>[2.9%] |
| Medium-run           | 2.568*<br>(1.423)<br>[10.5%] | 6.687*<br>(3.796)<br>[7.6%] | 8.097*<br>(4.589)<br>[7.3%] | 3.168**<br>(1.349)<br>[12.9%] | 6.659*<br>(3.652)<br>[7.5%] | 8.057*<br>(4.549)<br>[7.2%] |
| Mean (2010)          | 24.529                       | 88.232                      | 111.48                      | 24.529                        | 88.232                      | 111.48                      |
| R-Squared            | 0.766                        | 0.771                       | 0.772                       | 0.671                         | 0.712                       | 0.720                       |
| Observations         | 506                          | 506                         | 506                         | 4587                          | 4587                        | 4587                        |
| State of County FE   | Yes                          | Yes                         | Yes                         | Yes                           | Yes                         | Yes                         |
| Year FE              | Yes                          | Yes                         | Yes                         | Yes                           | Yes                         | Yes                         |
| $X_{st}$ or $X_{ct}$ | Yes                          | Yes                         | Yes                         | Yes                           | Yes                         | Yes                         |

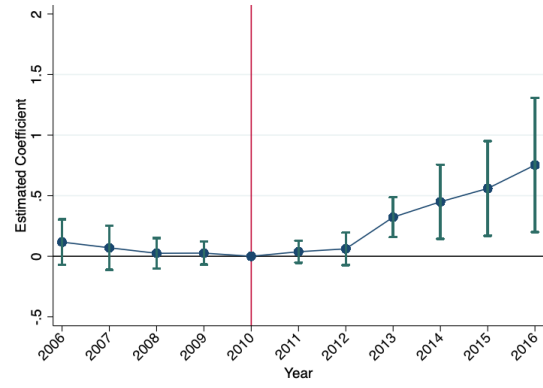
**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4, where The dependent variables are false negatives per 1,000 allegations by the duration between the placement decision and re-referral. Columns (1), (2), and (3) are state-level results whereas Columns (4), (5), and (6) are county-level results. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

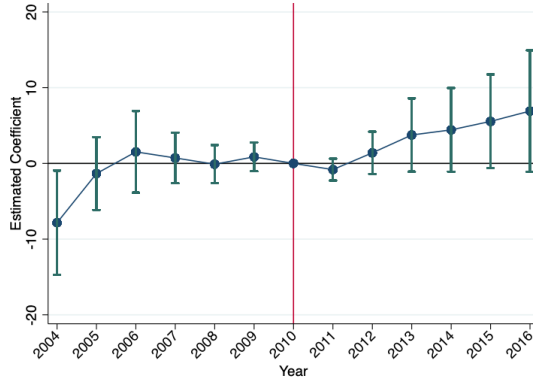
Figure A17: False Negative Rate Based on Service Provision



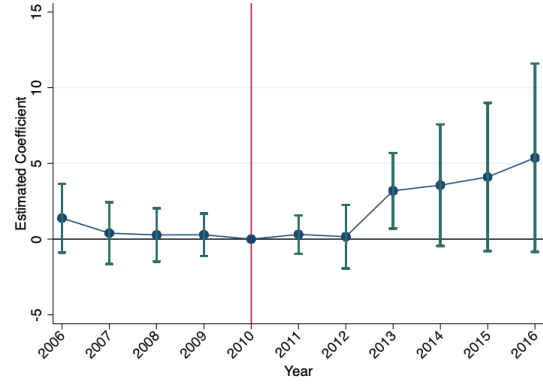
(a) FN by State, per 1,000 Children



(b) FN by County, per 1,000 Children



(c) FN by State, per 1,000 Allegations



(d) FN by County, per 1,000 Allegations

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Dependent variables are false negative rates based on service provision.

Table A16: Difference-in-Differences Results False Negative Rate Based on Service

|                      | (1)                           | (2)                            | (3)                          | (4)                          |
|----------------------|-------------------------------|--------------------------------|------------------------------|------------------------------|
|                      | FN                            | FN                             | FN                           | FN                           |
|                      | Children<br>State             | Children<br>County             | Allegations<br>State         | Allegations<br>County        |
| Pre-reformulation    | -0.025<br>(0.067)             | 0.058<br>(0.066)               | -0.964<br>(1.456)            | 1.006<br>(1.562)             |
| Short-run            | 0.131**<br>(0.063)<br>[12.7%] | 0.138***<br>(0.043)<br>[13.4%] | 1.356<br>(1.177)<br>[3.1%]   | 2.631**<br>(1.080)<br>[5.9%] |
| Medium-run           | 0.475**<br>(0.191)<br>[46.0%] | 0.579***<br>(0.205)<br>[56.1%] | 5.288*<br>(3.128)<br>[11.9%] | 7.413*<br>(4.173)<br>[16.7%] |
| Mean (2010)          | 1.032                         | 1.032                          | 44.332                       | 44.332                       |
| R-Squared            | 0.689                         | 0.679                          | 0.678                        | 0.691                        |
| Observations         | 492                           | 4558                           | 492                          | 4558                         |
| State or County FE   | Yes                           | Yes                            | Yes                          | Yes                          |
| Year FE              | Yes                           | Yes                            | Yes                          | Yes                          |
| $X_{st}$ or $X_{ct}$ | Yes                           | Yes                            | Yes                          | Yes                          |

**Notes.** This table reports point estimates and standard errors from Equation 3 and 4, where the dependent variables are false negatives per 1,000 allegations based on service provision. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$



## K Binned Treatment

Table A17: Difference-in-Differences Results for Binned Treatment

|                      | (1)<br>Allegations<br>Children<br>State | (2)<br>Allegations<br>Children<br>County | (3)<br>FN<br>Children<br>State | (4)<br>FN<br>Children<br>County | (5)<br>FN<br>Allegations<br>State | (6)<br>FN<br>Allegations<br>County |
|----------------------|-----------------------------------------|------------------------------------------|--------------------------------|---------------------------------|-----------------------------------|------------------------------------|
| PreQ2                | 0.803<br>(1.617)                        | 0.309<br>(0.625)                         | 0.028<br>(0.121)               | 0.049<br>(0.063)                | -3.267<br>(3.484)                 | -0.302<br>(1.708)                  |
| PreQ3                | 1.270<br>(2.476)                        | -1.208<br>(1.965)                        | -0.291<br>(0.208)              | -0.009<br>(0.096)               | -5.518<br>(3.359)                 | -0.056<br>(2.175)                  |
| PreQ4                | 0.039<br>(2.298)                        | 1.487<br>(2.764)                         | 0.055<br>(0.178)               | 0.061<br>(0.202)                | 1.779<br>(3.352)                  | 0.224<br>(3.400)                   |
| SrQ2                 | -2.065<br>(1.899)                       | 1.553**<br>(0.680)                       | 0.105<br>(0.158)               | 0.157<br>(0.099)                | 1.887<br>(2.667)                  | 1.074<br>(1.457)                   |
| SrQ3                 | 2.152<br>(2.650)                        | 4.441***<br>(1.222)                      | 0.184<br>(0.179)               | 0.274<br>(0.164)                | 0.614<br>(2.905)                  | 3.575<br>(2.846)                   |
| SrQ4                 | 2.026<br>(2.491)                        | 4.958**<br>(2.329)                       | 0.657**<br>(0.295)             | 0.385***<br>(0.134)             | 9.643*<br>(5.357)                 | 5.007*<br>(2.564)                  |
| MrQ2                 | -2.951<br>(3.066)                       | 3.387**<br>(1.373)                       | 0.163<br>(0.305)               | 0.327*<br>(0.170)               | 0.808<br>(4.778)                  | 2.334<br>(2.310)                   |
| MrQ3                 | 1.670<br>(3.799)                        | 6.606***<br>(2.403)                      | 0.229<br>(0.428)               | 0.872*<br>(0.442)               | -1.253<br>(5.211)                 | 7.763<br>(5.429)                   |
| MrQ4                 | 9.653*<br>(4.894)                       | 10.330***<br>(2.993)                     | 1.862**<br>(0.750)             | 1.361***<br>(0.447)             | 18.515*<br>(10.253)               | 13.365*<br>(6.975)                 |
| R-Squared            | 0.879                                   | 0.869                                    | 0.756                          | 0.712                           | 0.779                             | 0.705                              |
| Observations         | 634                                     | 7073                                     | 506                            | 4587                            | 506                               | 4587                               |
| State or County FE   | Yes                                     | Yes                                      | Yes                            | Yes                             | Yes                               | Yes                                |
| Year FE              | Yes                                     | Yes                                      | Yes                            | Yes                             | Yes                               | Yes                                |
| $X_{st}$ or $X_{ct}$ | Yes                                     | Yes                                      | Yes                            | Yes                             | Yes                               | Yes                                |

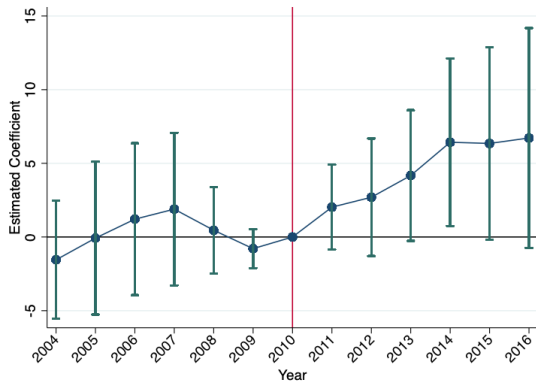
**Notes.** Standard errors in parentheses are clustered at the state level.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

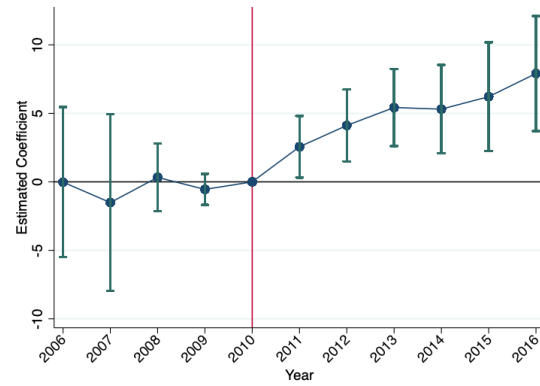


# L Binary Treatment

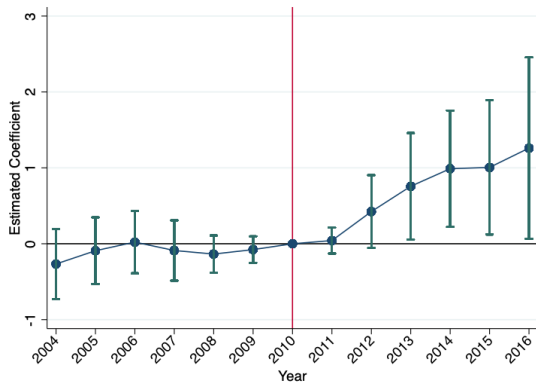
Figure A18: Event Study Results for Binary Treatment



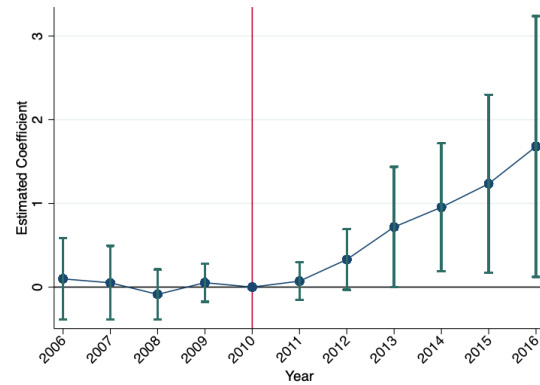
(a) Allegations by State



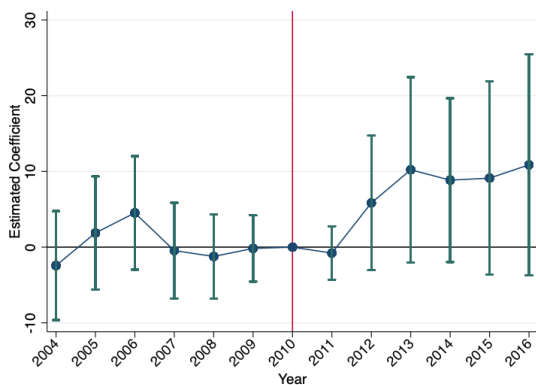
(b) Allegations by County



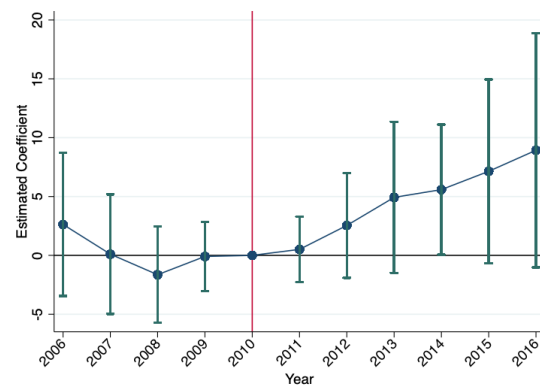
(c) FN by State, per 1,000 Children



(d) FN by County, per 1,000 Children



(e) FN by State, per 1,000 Allegations



(f) FN by County, per 1,000 Allegations

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Figures (a) and (b) report results for allegations. Figures (c), (d), (e), and (f) report results for false negatives. Figures (a), (c), and (e) are based on state-level analyses, whereas Figures (b), (d), and (f) are based on county-level analyses. Regressions are weighted by child population.

Table A18: Difference-in-Differences Results for Binary Treatment

|                      | (1)                          | (2)                            | (3)                           | (4)                           | (5)                         | (6)                          |
|----------------------|------------------------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|------------------------------|
|                      | Allegations                  | Allegations                    | FN                            | FN                            | FN                          | FN                           |
|                      | State                        | County                         | Children                      | Children                      | Allegations                 | Allegations                  |
|                      | State                        | County                         | State                         | County                        | State                       | County                       |
| Pre-reformulation    | 0.193<br>(1.739)             | -0.440<br>(1.752)              | -0.108<br>(0.151)             | -0.000<br>(0.099)             | 0.278<br>(2.668)            | 0.232<br>(1.953)             |
| Short-run            | 2.958*<br>(1.682)<br>[12.3%] | 4.018***<br>(1.191)<br>[16.8%] | 0.406**<br>(0.198)<br>[28.3%] | 0.239**<br>(0.110)<br>[16.6%] | 5.066<br>(3.712)<br>[8.2%]  | 3.516<br>(2.086)<br>[5.7%]   |
| Medium-run           | 6.480*<br>(3.270)<br>[27.0%] | 6.443***<br>(1.870)<br>[26.9%] | 1.075**<br>(0.465)<br>[74.8%] | 0.880**<br>(0.354)<br>[61.2%] | 9.466<br>(6.218)<br>[15.3%] | 8.427*<br>(4.380)<br>[13.6%] |
| Mean (2010)          | 23.976                       | 23.976                         | 1.437                         | 1.437                         | 61.804                      | 61.804                       |
| R-Squared            | 0.869                        | 0.867                          | 0.735                         | 0.709                         | 0.767                       | 0.704                        |
| Observations         | 634                          | 7073                           | 506                           | 4587                          | 506                         | 4587                         |
| State or County FE   | Yes                          | Yes                            | Yes                           | Yes                           | Yes                         | Yes                          |
| Year FE              | Yes                          | Yes                            | Yes                           | Yes                           | Yes                         | Yes                          |
| $X_{st}$ or $X_{ct}$ | Yes                          | Yes                            | Yes                           | Yes                           | Yes                         | Yes                          |

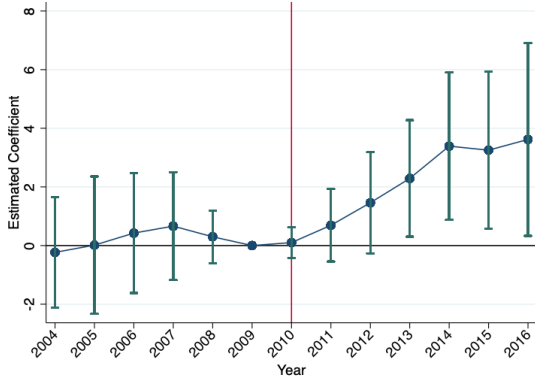
**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4. Columns (1) and (2) report results for allegations. Columns (3), (4), (5), and (6) report results for false negatives. Figures (1), (3), and (5) are based on state-level analyses, whereas Figures (2), (4), and (6) are based on county-level analyses. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

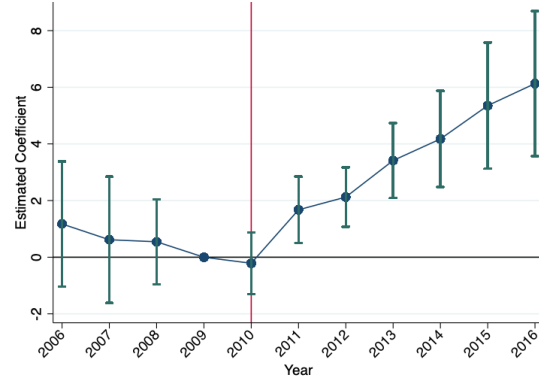


# M Results for Year 2009 Normalized

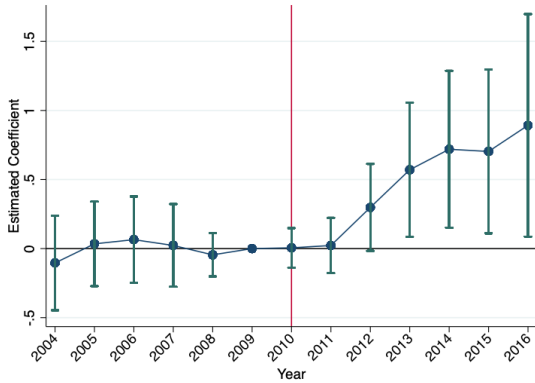
Figure A19: Event Study Results for 2009 Normalized



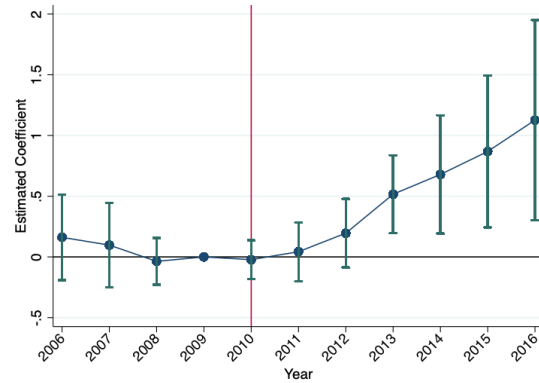
(a) Professional Allegations by State



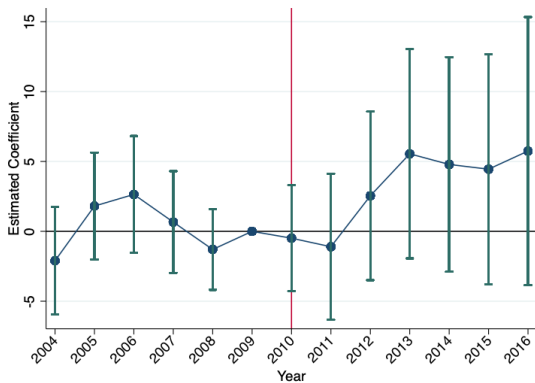
(b) Professional Allegations by County



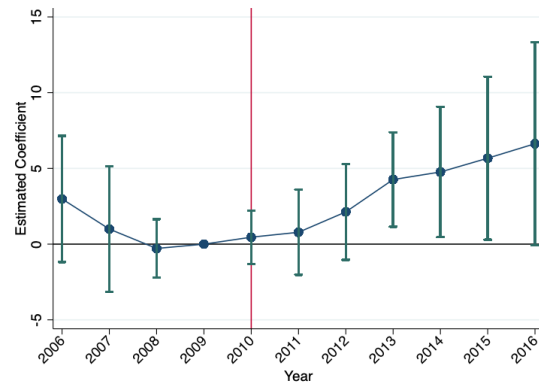
(c) FN by State, per 1,000 Children



(d) FN by County, per 1,000 Children



(e) FN by State, per 1,000 Allegations



(f) FN by County, per 1,000 Allegations

**Notes.** Each figure reports point estimates and 95 percent confidence intervals on  $\delta_t$  (normalized to 0 in 2010) from Equation 1 and Equation 2 that are adjusted for within-state clustering. Figures (a) and (b) report results for allegations. Figures (c), (d), (e), and (f) report results for false negatives. Figures (a), (c), and (e) are based on state-level analyses, whereas Figures (b), (d), and (f) are based on county-level analyses. Regressions are weighted by child population.

Table A19: Difference-in-Differences Results for 2009 Normalized

|                      | (1)                           | (2)                            | (3)                           | (4)                            | (5)                         | (6)                          |
|----------------------|-------------------------------|--------------------------------|-------------------------------|--------------------------------|-----------------------------|------------------------------|
|                      | Allegations                   | Allegations                    | FN                            | FN                             | FN                          | FN                           |
|                      | State                         | County                         | Children                      | Children                       | Allegations                 | Allegations                  |
|                      | State                         | County                         | State                         | County                         | State                       | County                       |
| Pre-reformulation    | 0.248<br>(0.822)              | 0.777<br>(0.932)               | -0.015<br>(0.072)             | 0.049<br>(0.081)               | 0.055<br>(1.310)            | 0.676<br>(1.435)             |
| Short-run            | 0.746*<br>(0.442)<br>[3.2%]   | 1.203***<br>(0.338)<br>[5.1%]  | 0.209**<br>(0.091)<br>[14.7%] | 0.181***<br>(0.054)<br>[12.7%] | 2.780*<br>(1.629)<br>[4.5%] | 2.542**<br>(1.130)<br>[4.1%] |
| Medium-run           | 3.095**<br>(1.231)<br>[13.1%] | 4.701***<br>(0.896)<br>[19.9%] | 0.545**<br>(0.220)<br>[38.4%] | 0.626***<br>(0.197)<br>[44.1%] | 5.002*<br>(2.936)<br>[8.0%] | 5.998**<br>(2.776)<br>[9.6%] |
| Mean (2009)          | 23.661                        | 23.661                         | 1.42                          | 1.42                           | 62.389                      | 62.389                       |
| R-Squared            | 0.869                         | 0.866                          | 0.738                         | 0.709                          | 0.771                       | 0.703                        |
| Observations         | 634                           | 7073                           | 506                           | 4587                           | 506                         | 4587                         |
| State of County FE   | Yes                           | Yes                            | Yes                           | Yes                            | Yes                         | Yes                          |
| Year FE              | Yes                           | Yes                            | Yes                           | Yes                            | Yes                         | Yes                          |
| $X_{st}$ or $X_{ct}$ | Yes                           | Yes                            | Yes                           | Yes                            | Yes                         | Yes                          |

**Notes.** This table reports point estimates and standard errors from Equation 3 and Equation 4. Columns (1) and (2) report results for allegations. Columns (3), (4), (5), and (6) report results for false negatives. Figures (1), (3), and (5) are based on state-level analyses, whereas Figures (2), (4), and (6) are based on county-level analyses. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

# N Heterogeneity Analysis for Placements - State

Table A20: Placement Rate by Type and Child Characteristics - State

|                   | (1)                          | (2)                        | (3)                          | (4)                          | (5)                          | (6)                          | (7)                          | (8)                          |
|-------------------|------------------------------|----------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|                   | Neglect                      | Physical                   | White                        | Black                        | Male                         | Female                       | Young                        | Old                          |
| Pre-reformulation | -3.212<br>(3.527)            | -1.094<br>(2.714)          | -0.703<br>(1.762)            | 1.897<br>(2.320)             | -0.270<br>(1.586)            | 0.109<br>(1.794)             | -0.429<br>(1.808)            | 0.522<br>(1.723)             |
| Short-run         | -2.403<br>(2.563)<br>[-2.4%] | 1.886<br>(2.840)<br>[3.3%] | 0.011<br>(1.846)<br>[0.0%]   | 0.391<br>(2.623)<br>[0.5%]   | -0.936<br>(1.700)<br>[-1.4%] | -0.366<br>(1.682)<br>[-0.5%] | -0.007<br>(1.858)<br>[0.0%]  | -0.859<br>(1.625)<br>[-1.7%] |
| Medium-run        | -2.208<br>(3.886)<br>[-2.2%] | 2.946<br>(3.154)<br>[5.1%] | -2.861<br>(3.511)<br>[-4.2%] | -2.159<br>(3.679)<br>[-2.9%] | -2.500<br>(2.945)<br>[-3.8%] | -1.409<br>(2.858)<br>[-2.1%] | -1.013<br>(3.246)<br>[-1.2%] | -2.034<br>(2.817)<br>[-3.9%] |
| Mean (2010)       | 99.107                       | 58.72                      | 68.378                       | 74.494                       | 66.371                       | 66.677                       | 84.102                       | 51.854                       |
| R-Squared         | 0.871                        | 0.829                      | 0.769                        | 0.779                        | 0.741                        | 0.751                        | 0.753                        | 0.749                        |
| Observations      | 506                          | 506                        | 506                          | 506                          | 506                          | 506                          | 506                          | 506                          |
| State FE          | Yes                          | Yes                        | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          |
| Year FE           | Yes                          | Yes                        | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          |
| $X_{st}$          | Yes                          | Yes                        | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          |

**Notes.** This table reports point estimates and standard errors from Equation 3, where the dependent variables are foster care placements per 1,000 allegations, broken down by child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$



## O Heterogeneity Analysis for Placements - County

Table A21: Placement Rate by Type and Child Characteristics - County

|                   | (1)                          | (2)                        | (3)                          | (4)                          | (5)                          | (6)                          | (7)                          | (8)                          |
|-------------------|------------------------------|----------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|------------------------------|
|                   | Neglect                      | Physical                   | White                        | Black                        | Male                         | Female                       | Young                        | Old                          |
| Pre-reformulation | -6.517<br>(4.526)            | -1.359<br>(3.337)          | -2.999<br>(2.387)            | -2.097<br>(3.077)            | -2.277<br>(2.650)            | -0.988<br>(2.547)            | -4.215<br>(2.759)            | 0.037<br>(2.468)             |
| Short-run         | 1.137<br>(1.998)<br>[1.1%]   | 3.885<br>(2.731)<br>[6.6%] | 3.580<br>(2.395)<br>[5.2%]   | 1.771<br>(2.468)<br>[2.4%]   | -0.015<br>(1.508)<br>[0.0%]  | 1.198<br>(1.559)<br>[1.8%]   | 0.912<br>(1.705)<br>[1.1%]   | 0.670<br>(1.435)<br>[1.3%]   |
| Medium-run        | -3.142<br>(4.675)<br>[-3.2%] | 1.776<br>(4.134)<br>[3.0%] | -0.078<br>(4.575)<br>[-0.1%] | -0.070<br>(5.329)<br>[-0.1%] | -1.996<br>(3.501)<br>[-3.0%] | -0.325<br>(3.400)<br>[-0.5%] | -1.456<br>(3.424)<br>[-1.7%] | -0.462<br>(3.492)<br>[-0.9%] |
| Mean (2010)       | 99.107                       | 58.72                      | 68.378                       | 74.494                       | 66.371                       | 66.677                       | 84.102                       | 51.854                       |
| R-Squared         | 0.820                        | 0.784                      | 0.755                        | 0.593                        | 0.739                        | 0.749                        | 0.749                        | 0.735                        |
| Observations      | 5544                         | 5544                       | 5544                         | 5544                         | 5544                         | 5544                         | 5544                         | 5544                         |
| County FE         | Yes                          | Yes                        | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          |
| Year FE           | Yes                          | Yes                        | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          |
| $X_{ct}$          | Yes                          | Yes                        | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          | Yes                          |

**Notes.** This table reports point estimates and standard errors from Equation 4, where the dependent variables are foster care placements per 1,000 allegations, broken down by child characteristics. Regressions are weighted by child population. Standard errors in parentheses are clustered at the state level. Percentage changes from the baseline mean in 2010 are reported in square brackets.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## P Policy Variables

Table A22: Must-Access PDMPs and Medicaid Expansion

| Year | PDMP           | Medicaid                                                               |
|------|----------------|------------------------------------------------------------------------|
| 2006 | NV             |                                                                        |
| 2010 | OK             |                                                                        |
| 2011 | OH             |                                                                        |
| 2012 | DE, KY, NM, WV |                                                                        |
| 2013 | MA, NY, TN, VT |                                                                        |
| 2014 | IN, LA         | AZ, AR, CA, CO, CT, HI, IL, IA, KY, MD, MI, NV, NH, NJ, OH, RI, WA, WV |
| 2015 | CT, NJ, VA     | PA, IN, AK                                                             |
| 2016 | NH, RI         | MT, LA                                                                 |

**Notes.** Must-access PDMP implementation dates are based on Sacks et al. (2021) with five corrections made by Evans et al. (2021).